

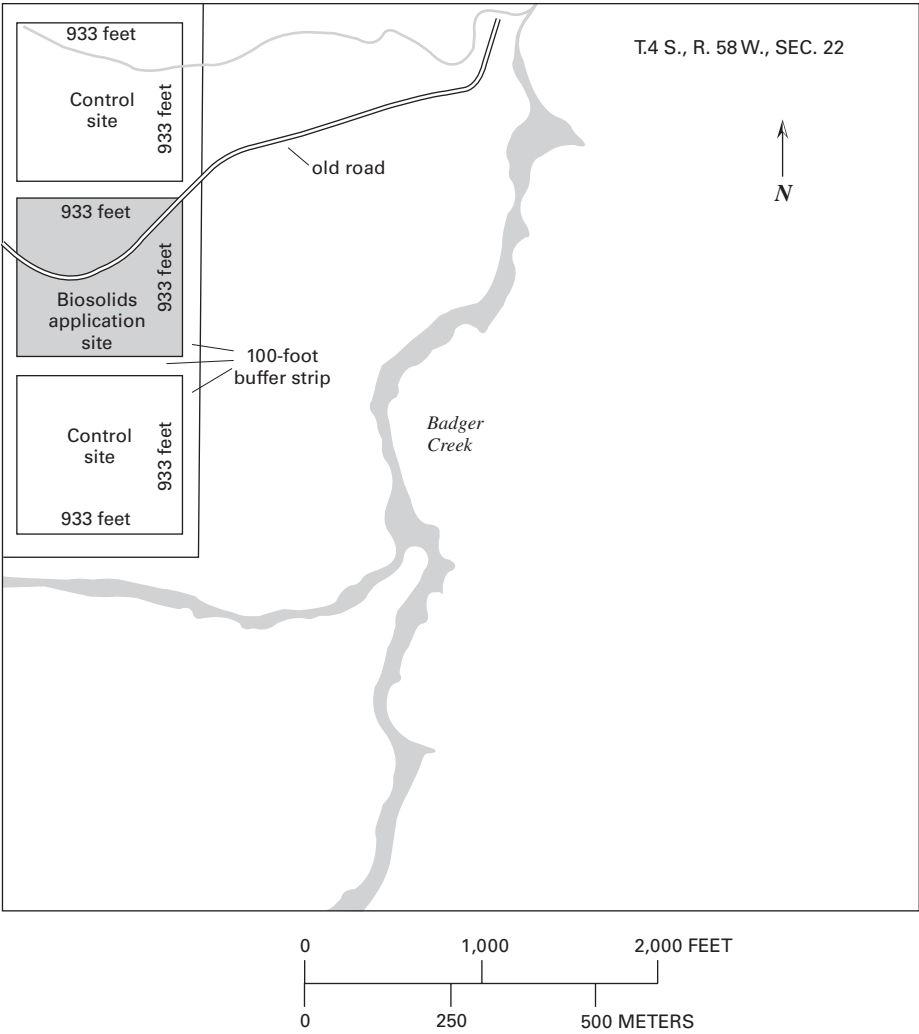


Figure 3. Arapahoe County, Colorado, soil-monitoring site: T. 4 S., R. 58 W., sec. 22 (from Metro Wastewater Reclamation District).

22 **Data for a Biosolids-Application Area near Deer Trail, Colorado, 2000**

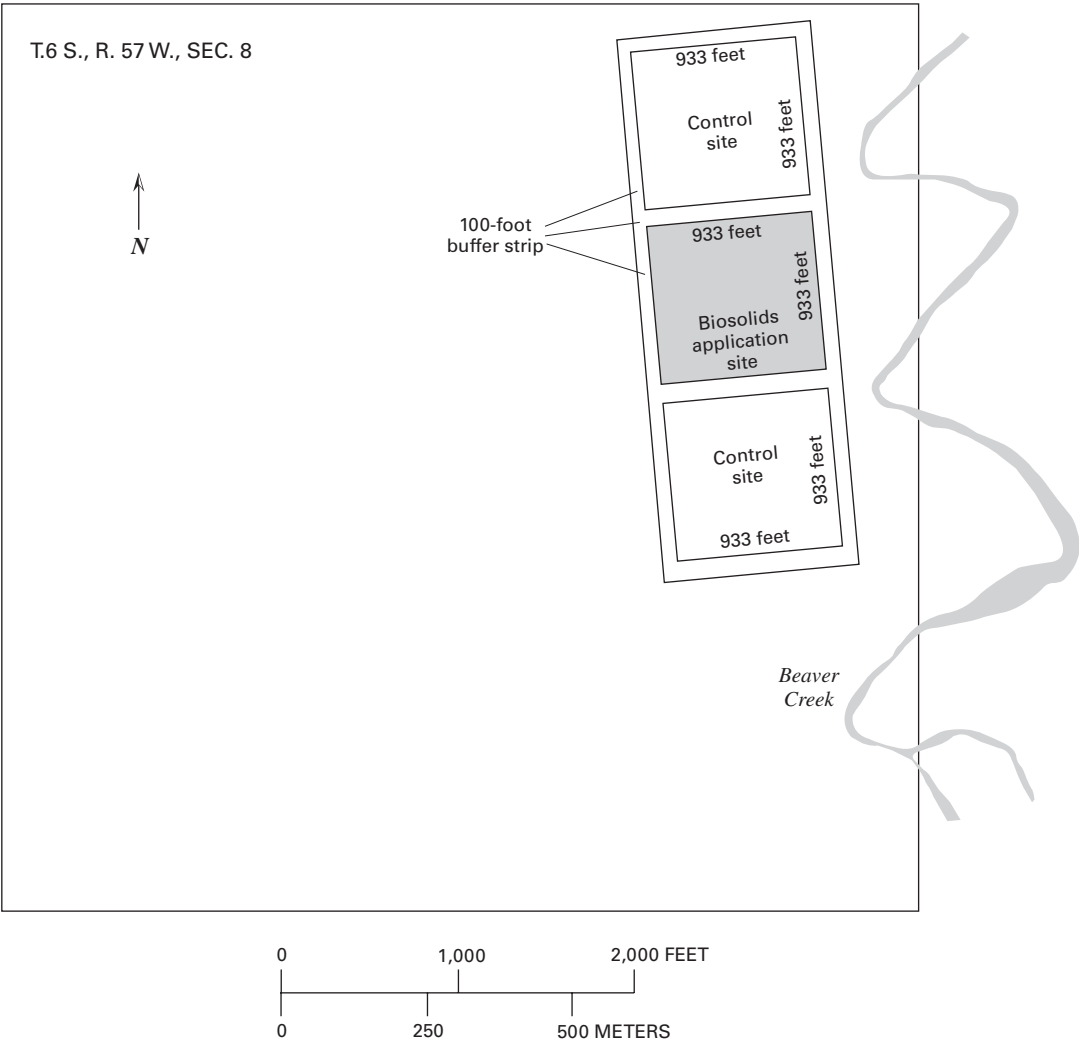


Figure 4. Elbert County, Colorado, soil-monitoring site: T. 6 S., R. 57 W., sec. 8 (from Metro Wastewater Reclamation District)..

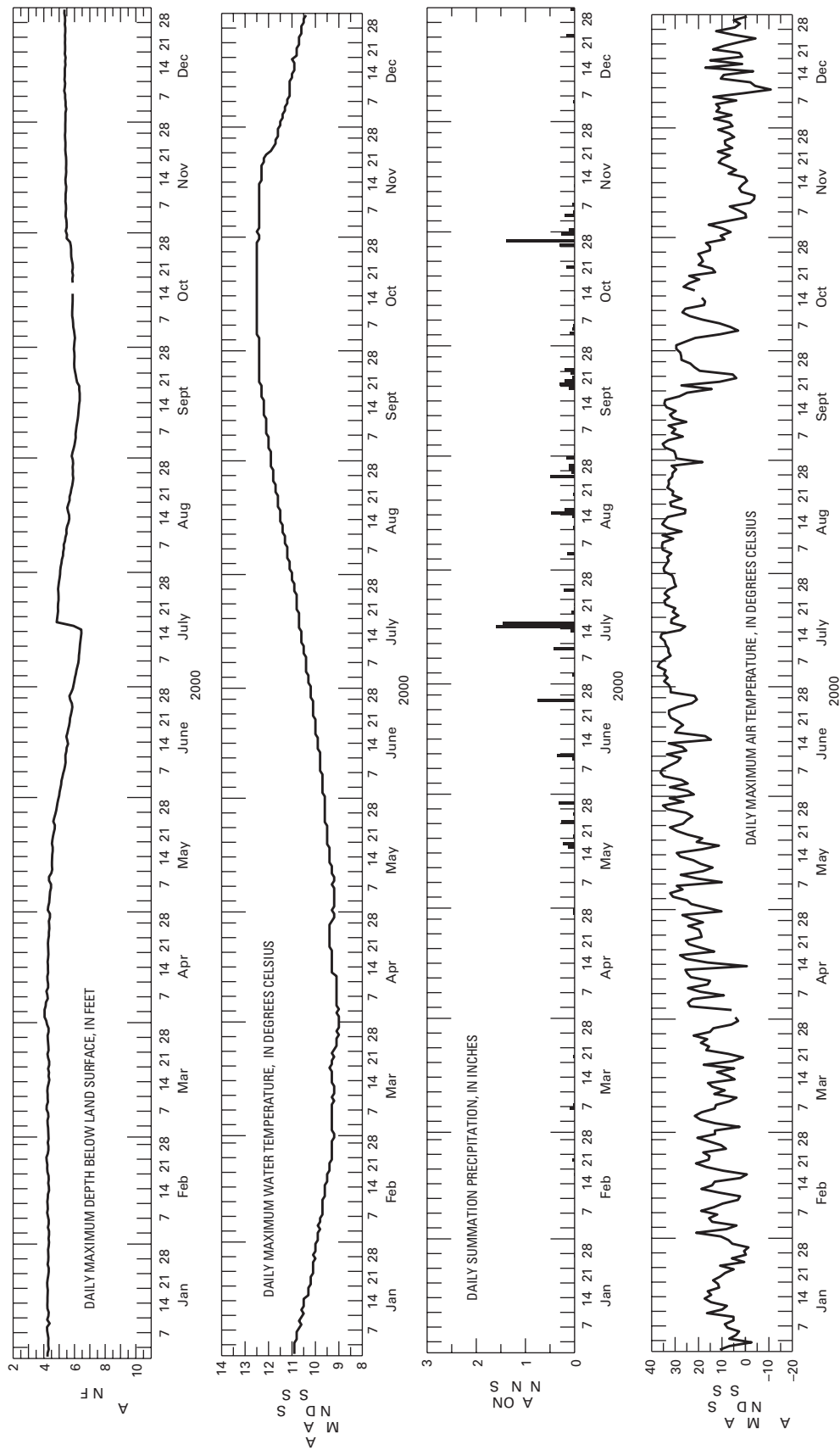


Figure 7. Continuous water-level, water-temperature, precipitation, and air-temperature data for well DTX2 near Deer Trail, Colorado, 2000

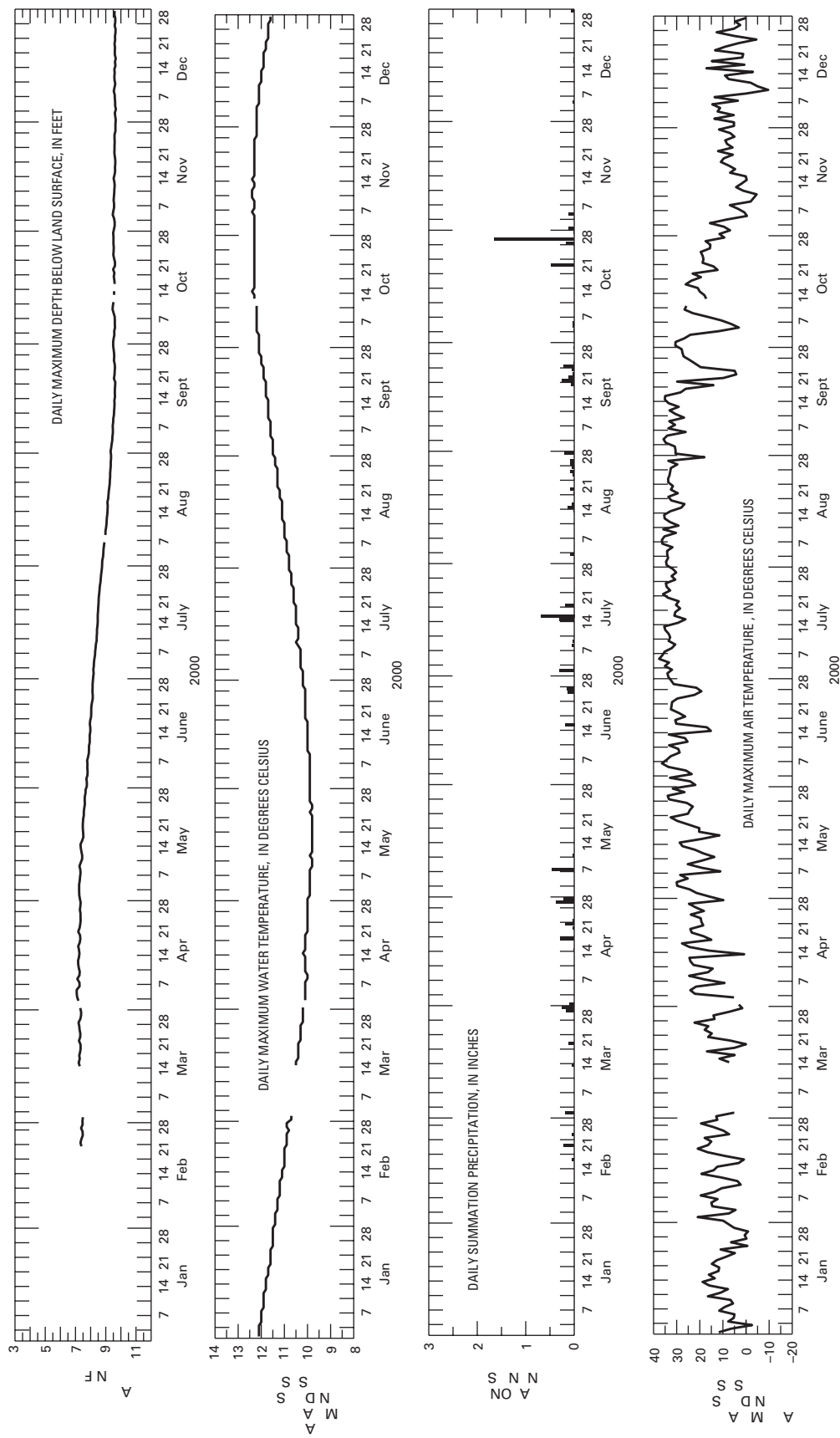


Figure 8. Continuous water-level, water-temperature, precipitation, and air-temperature data for well DTX5 near Deer Trail, Colorado, 2000

28 Data for a Biosolids-Application Area near Deer Trail, Colorado, 2000

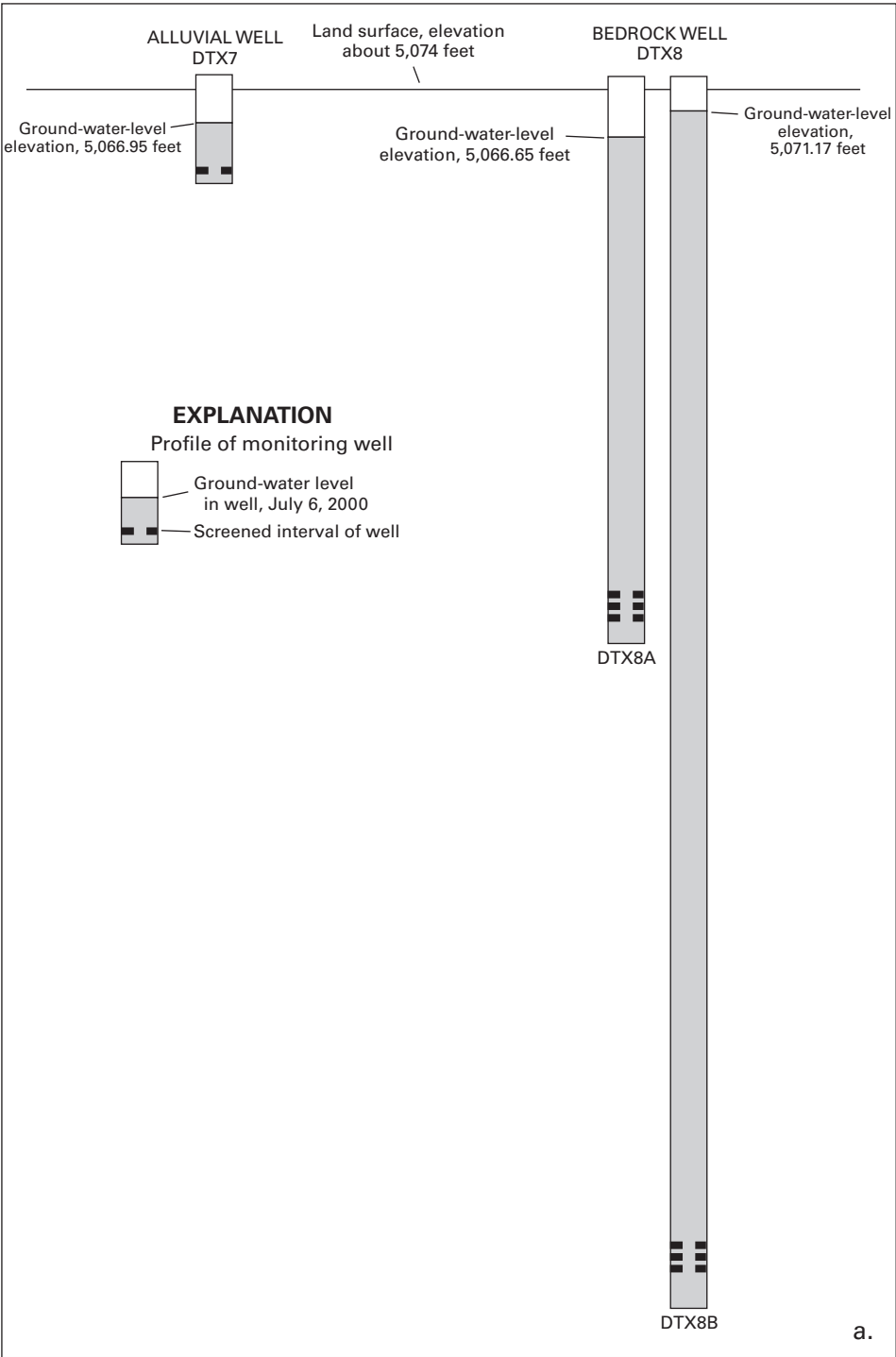


Figure 10. Water levels for the recharge-evaluation site containing well DTX7 and nested well DTX8 (includes DTX8A and DTX8B) near Deer Trail, Colorado, for (a) July, (b) August, and (c) September 2000. (Elevation is in feet above NAVD 88.)

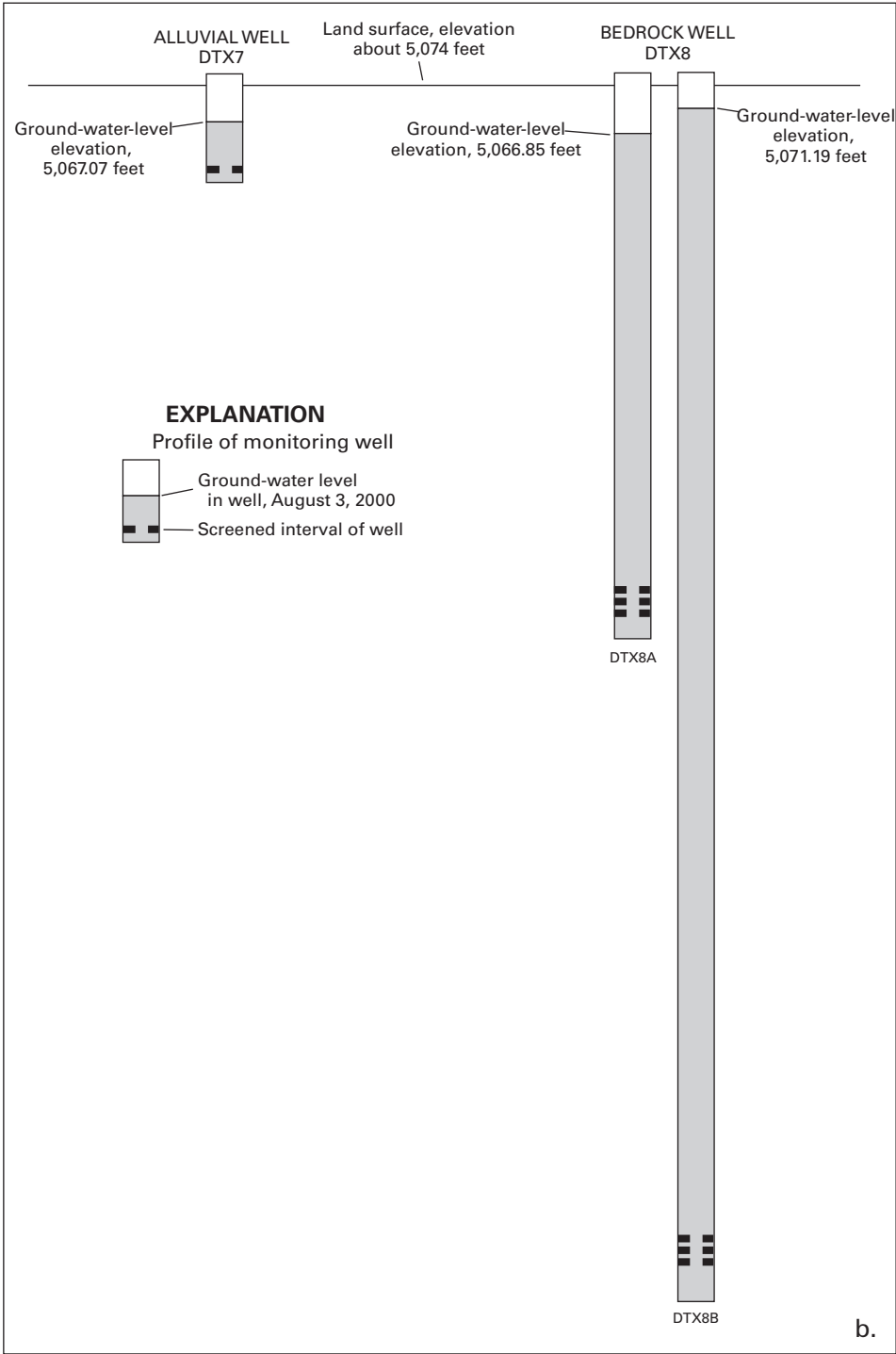


Figure 10. Water levels for the recharge-evaluation site containing well DTX7 and nested well DTX8 (includes DTX8A and DTX8B) near Deer Trail, Colorado, for (a) July, (b) August, and (c) September 2000. (Elevation is in feet above NAVD 88.)—Continued.

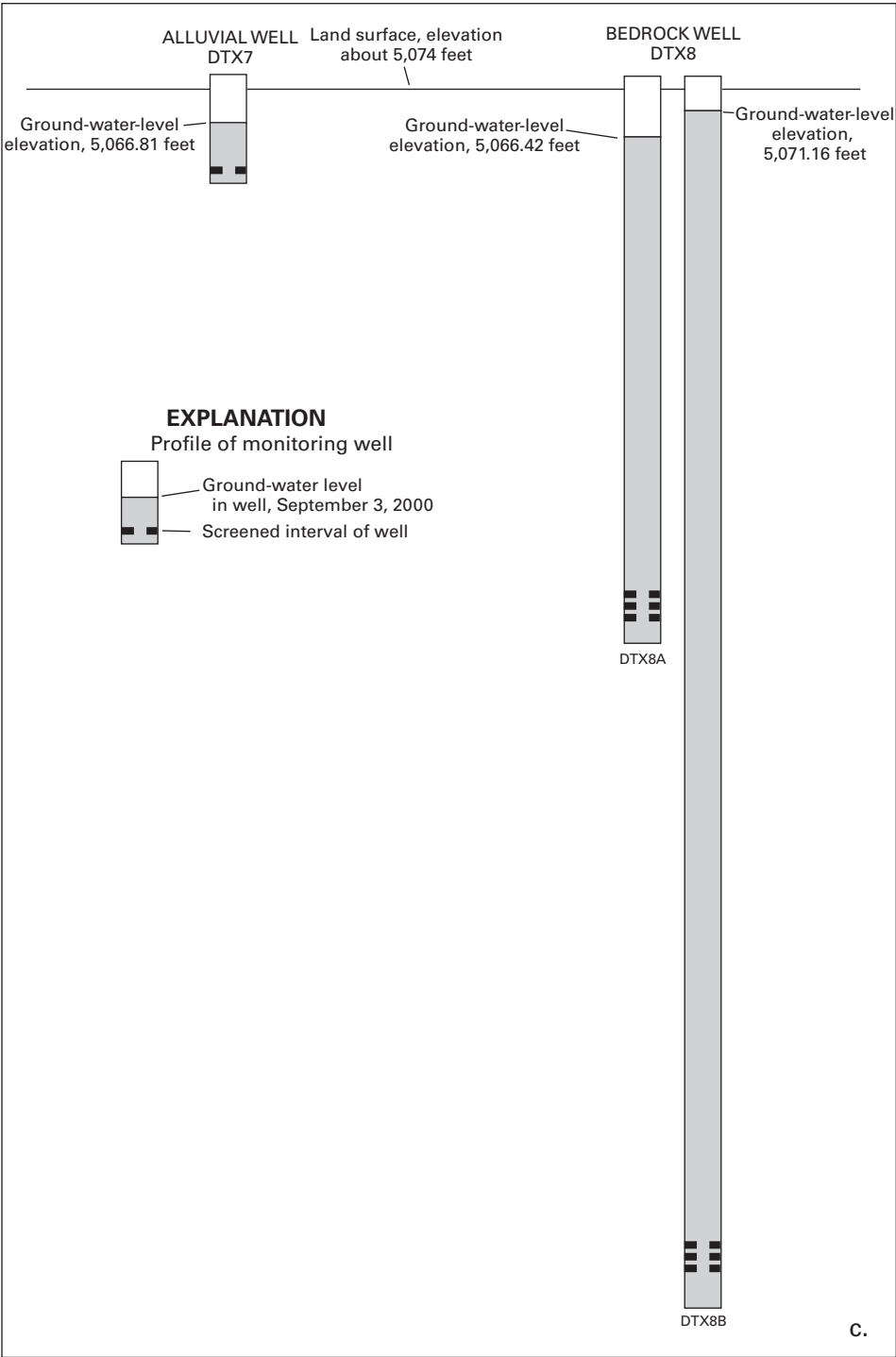
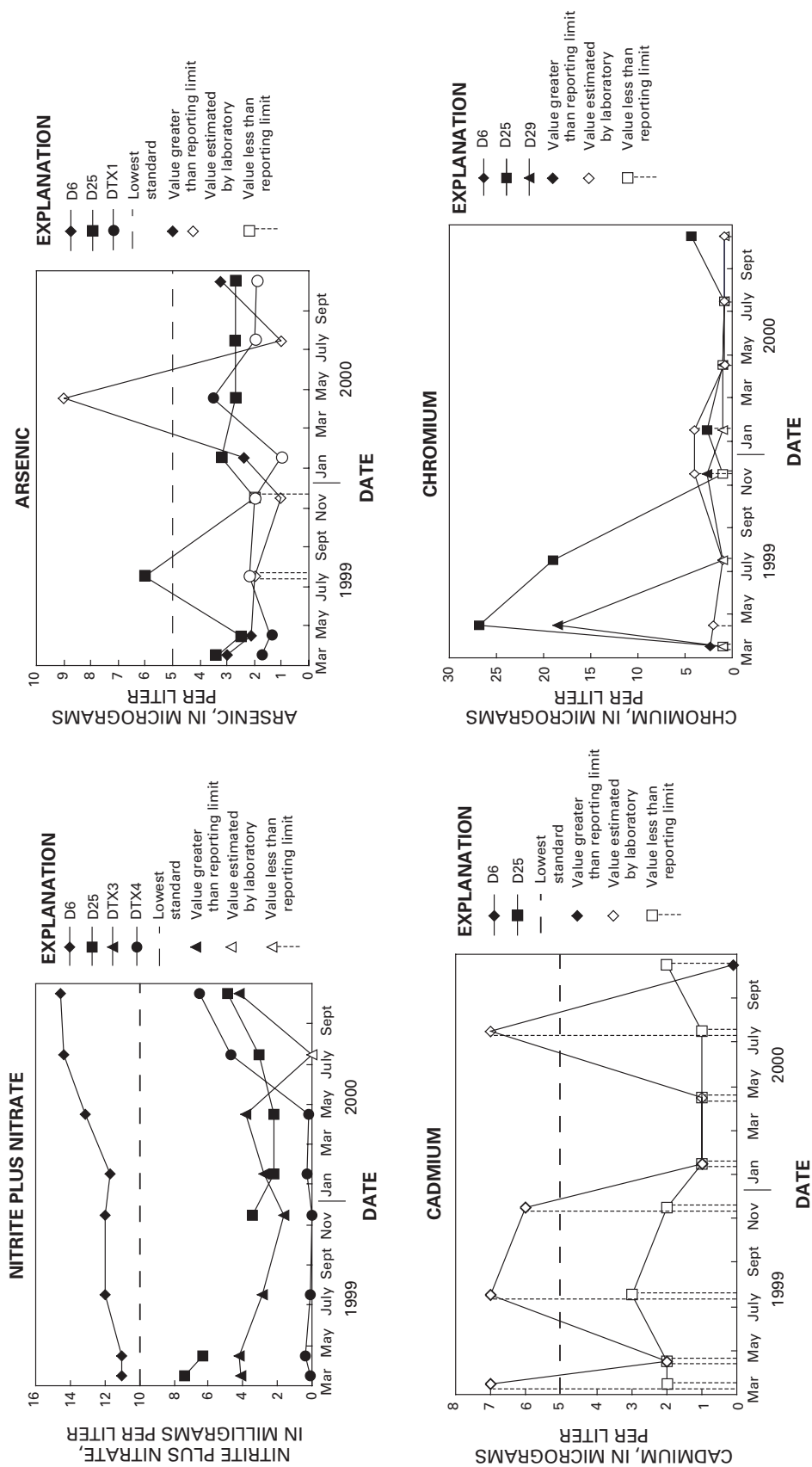


Figure 10. Water levels for the recharge-evaluation site containing well DTX7 and nested well DTX8 (includes DTX8A and DTX8B) near Deer Trail, Colorado, for (a) July, (b) August, and (c) September 2000. (Elevation is in feet above NAVD 88.)—Continued.



36 Data for a Biosolids-Application Area near Deer Trail, Colorado, 2000

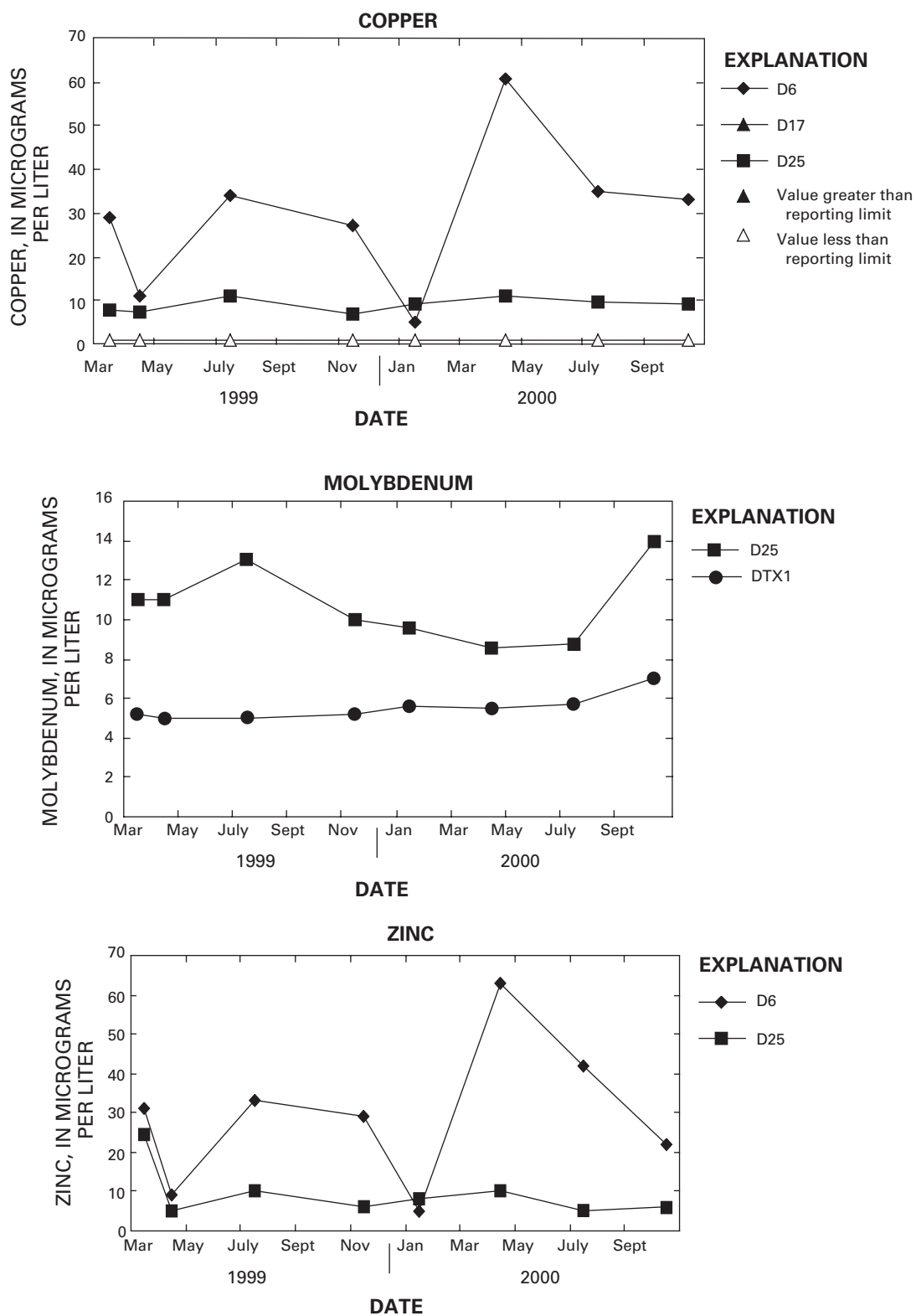


Figure 13. Ground-water concentrations near Deer Trail, Colorado, for selected constituents and selected wells, 1999-2000—Continued.

TERMS AND ABBREVIATIONS

The following terms and abbreviations are used in tables 1–19:

bls	below land surface
bmp	below the measuring point of the well casing
mi	mile
ft	feet
hhmm	24-hour time
mm/dd/yy	numerical date format for two-digit month/two-digit day/ and the last two digits of the year
C	Celsius
cm	centimeter
°	degree
in. or "	inch
μS/cm	microsiemens per centimeter at 25°Celsius
pH units	are the negative base-10 log of the hydrogen-ion activity in moles per liter
mg/L	milligrams per liter
μg/L	micrograms per liter
pCi/L	picocuries per liter
dissolved	refers to that fraction of material in a water sample that passes through a 0.45-μm membrane filter
>	greater than
<	less than
g/kg	grams per kilogram
mg/kg	milligrams per kilogram
pCi/g	picocuries per gram
μg/g	micrograms per gram
NAD 83	North American Datum 1983
NAVD 88	North American Vertical Datum 1988

Table 1. Biosolids applications near Deer Trail, Colorado, 2000

[All information provided by Metro Wastewater Reclamation District; DC, destination code (shown in fig. 2); legal description is of the form *quarter-section section township range*; Ha, hectares; CAKE, Grade I Class B biosolids; MAC, biosolids amended with wood fiber; dT, dry tons; dMT, dry metric tons; lbs/Acre, pounds per acre; Kg/Ha, kilograms per hectare; Y, yes; N, No; O/GR, oats and grass; -, no data]

DC	Legal description	Area applied to		County	Biosolids product applied	Start date	Stop date	Total loads	Total wet tons	Total dry tons		Loading rate, tons per acre		Nitrogen loading rate		Crop	Reclamation project
		Acres	Ha							dT	dMT	Cake dT	dMT	lbs/Acre	Kg/Ha		
300	N 1/2 SEC 15 T5S R58W	160	64.8	ARAPAHOE	CAKE	7/18/00	7/21/00	82	1,835	299	271.25	1.87	4.19	46	51.57	WHEAT	N
300	N 1/2 SEC 15 T5S R58W	160	64.8	ARAPAHOE	CAKE	9/18/00	11/18/00	51	1,121	180	163.30	1.13	2.53	28	31.39	WHEAT	N
301	S 1/2 SEC 15 T5S R58W	155.7	63.0	ARAPAHOE	CAKE	7/22/00	7/25/00	95	2,125	329	298.47	2.11	4.73	52	58.30	WHEAT	N
301	S 1/2 SEC 15 T5S R58W	78	31.6	ARAPAHOE	CAKE/MAC	9/19/00	11/18/00	45	999	165	149.69	1.92	4.30	50	56.06	WHEAT	N
303	S 1/2 SEC 16 T5S R58W	148	59.9	ARAPAHOE	CAKE/MAC	7/25/00	7/30/00	118	2,628	424	384.65	2.77	6.21	70	78.48	WHEAT	N
306	N 1/2 SEC 20 T5S R58W	107.3	43.4	ARAPAHOE	CAKE	3/10/00	3/17/00	45	1,028	164	148.78	1.53	3.43	43	48.21	WHEAT	N
307	S 1/2 SEC 20 T5S R58W	107.1	43.3	ARAPAHOE	CAKE	3/11/00	6/21/00	80	1,824	279	253.11	2.61	5.85	69	77.36	WHEAT	N
308	N 1/2 SEC 21 T5S R58W	156.6	63.4	ARAPAHOE	CAKE/MAC	11/22/00	11/29/00	93	2,090	350	317.52	2.04	4.57	58	65.02	WHEAT	N
309	S 1/2 SEC 21 T5S R58W	129.3	52.3	ARAPAHOE	CAKE	11/23/00	11/28/00	75	1,707	268	243.13	2.07	4.64	56	62.78	WHEAT	N
310	N 1/2 SEC 22 T5S R58W	69.3	28.0	ARAPAHOE	CAKE	11/19/00	11/22/00	54	1,219	274	248.57	2.74	6.14	67	75.11	WHEAT	N
311	S 1/2 SEC 22 T5S R58W	66.1	26.8	ARAPAHOE	CAKE/MAC	11/18/00	11/21/00	48	1,081	180	163.30	2.50	5.61	66	73.99	WHEAT	N
312	N 1/2 SEC 28 T5S R58W	166.3	67.3	ARAPAHOE	CAKE/MAC	11/29/00	12/5/00	135	2,970	510	462.67	2.72	6.10	83	93.05	WHEAT	N
313	S 1/2 SEC 28 T5S R58W	224.7	90.9	ARAPAHOE	CAKE	11/30/00	12/8/00	71	1,585	251	227.71	1.12	2.51	32	35.88	WHEAT	N
315	S 1/2 SEC 29 T5S R58W	23.9	9.7	ARAPAHOE	CAKE	12/8/00	12/9/00	18	416	67	60.78	2.80	6.28	76	85.20	WHEAT	N
317	S 1/2 SEC 32 T5S R58W	159.6	64.6	ARAPAHOE	CAKE/MAC	6/9/99	3/4/00	722	16,122	2,674	2,425.85	16.30	36.55	415	465.26	GRASS	Y
318	N 1/2 SEC 33 T5S R58W	145.3	58.8	ARAPAHOE	CAKE/MAC	12/11/00	12/17/00	116	2,490	278	252.20	2.68	6.01	80	89.69	WHEAT	N
319	S 1/2 SEC 33 T5S R58W	94.3	38.2	ARAPAHOE	CAKE/MAC	7/9/99	2/28/00	137	3,094	519	470.84	5.20	11.66	130	145.74	O/GR	Y

Table 1. Biosolids applications near Deer Trail, Colorado, 2000—Continued

[All information provided by Metro Wastewater Reclamation District; DC, destination code (shown in fig. 2); legal description is of the form *quarter-section section township range*; Ha, hectares; CAKE, Grade I Class B biosolids; MAC, biosolids amended with wood fiber; dT, dry tons; dMT, dry metric tons; lbs/Acre, pounds per acre; Kg/Ha, kilograms per hectare; Y, yes; N, No; O/GR, oats and grass; --, no data]

DC	Legal description	Area applied to		County	Biosolids product applied	Start date	Stop date	Total loads	Total wet tons	Total dry tons		Loading rate, tons per acre		Nitrogen loading rate		Crop	Reclamation project
		Acres	Ha							dT	dMT	Cake dT	dMT	lbs/Acre	Kg/Ha		
319	S 1/2 SEC 33 T5S R58W	50.2	20.3	ARAPAHOE	CAKE/MAC	12/9/00	12/11/00	41	902	157	142.43	2.83	6.35	81	90.81	WHEAT	N
322	N 1/2 SEC 3 T6S R58W	215.7	87.3	ELBERT	CAKE	12/19/00	12/26/00	121	2,760	447	405.52	2.07	4.64	65	72.87	WHEAT	N
323	S 1/2 SEC 3 T6S R58W	230.6	93.3	ELBERT	CAKE	12/26/00	01/04/01	184	4,143	683	619.62	2.96	6.64	--	--	--	--
324	N 1/2 SEC 4 T6S R58W	38	15.4	ELBERT	CAKE/MAC	12/17/00	12/18/00	29	649	117	106.14	2.68	6.01	84	94.17	WHEAT	N
325	S 1/2 SEC 4 T6S R58W	38	15.4	ELBERT	CAKE/MAC	12/16/00	12/20/00	31	708	124	112.49	2.87	6.43	89	99.78	WHEAT	N
340	W 1/2 SEC 9 T4S R58W	258.4	104.6	ARAPAHOE	CAKE	10/06/00	06/20/01	99	2,207	487	441.81	1.88	4.22	--	--	--	--
341	N 1/2 SEC 8 T4S R58W	300	121.4	ARAPAHOE	CAKE/MAC	8/29/00	11/17/00	231	5,018	810	734.83	2.60	5.83	69	77.36	WHEAT	N
342	S 1/2 SEC 8 T4S R58W	311.3	126.0	ARAPAHOE	CAKE/MAC	8/22/00	11/13/00	217	4,763	828	751.16	2.49	5.58	66	73.99	WHEAT	N
343	N 1/2 SEC 17 T4S R58W	327.4	132.5	ARAPAHOE	CAKE/MAC	8/18/00	11/10/00	259	5,777	984	892.68	2.91	6.52	81	90.81	WHEAT	N
344	S 1/2 SEC 17 T4S R58W	280	113.3	ARAPAHOE	CAKE/MAC	8/8/00	8/18/00	223	5,125	855	775.66	2.91	6.52	75	84.08	WHEAT	N
345	N 1/2 SEC 21 T4S R58W	310	125.5	ARAPAHOE	CAKE/MAC	7/29/00	11/5/00	255	5,823	975	884.52	2.94	6.59	78	87.45	WHEAT	N
346	S 1/2 SEC 21 T4S R58W	300	121.4	ARAPAHOE	CAKE/MAC	7/31/00	10/26/00	224	5,000	831	753.88	2.67	5.99	69	77.36	WHEAT	N
351	N 1/2 SEC 28 T4S R58W	290	117.4	ARAPAHOE	CAKE/MAC	8/27/00	9/16/00	210	4,591	743	674.05	2.38	5.34	59	66.14	WHEAT	N
352	S 1/2 SEC 28 T4S R58W	299	121.0	ARAPAHOE	CAKE	9/2/00	9/9/00	211	4,674	749	679.49	2.51	5.63	61	68.39	WHEAT	N
412	N 1/2 SEC 1 T6S R57W	343	138.8	ELBERT	CAKE	9/26/00	10/2/00	140	3,117	480	435.46	1.40	3.14	40	44.84	WHEAT	N
413	S 1/2 SEC 1 T6S R57W	343	138.8	ELBERT	CAKE/MAC	10/2/00	10/15/00	248	5,465	883	801.06	2.43	5.45	66	73.99	WHEAT	N
425	S 1/2 SEC 11 T6S R57W	20	8.1	ELBERT	CAKE	1/15/00	1/15/00	12	268	39	35.38	1.95	4.37	53	59.42	WHEAT	N
448	N 1/2 SEC 21 T6S R57W	223	90.2	ELBERT	CAKE/MAC	2/10/00	3/26/00	178	3,984	655	594.22	2.81	6.30	77	86.32	WHEAT	N
449	S 1/2 SEC 21 T6S R57W	123.7	50.1	ELBERT	CAKE/MAC	3/4/00	3/30/00	100	2,237	365	331.13	2.74	6.14	76	85.20	WHEAT	N

40 Data for a Biosolids-Application Area near Deer Trail, Colorado, 2000

Table 1. Biosolids applications near Deer Trail, Colorado, 2000—Continued

[All information provided by Metro Wastewater Reclamation District; DC, destination code (shown in fig. 2); legal description is of the form *quarter-section township range*; Ha, hectares; CAKE, Grade I Class B biosolids; MAC, biosolids amended with wood fiber; dT, dry tons; dMT, dry metric tons; lbs/Acre, pounds per acre; Kg/Ha, kilograms per hectare; Y, yes; N, No; O/GR, oats and grass; --, no data]

DC	Legal description	Area applied to		County	Biosolids product applied	Start date	Stop date	Total loads	Total wet tons	Total dry tons		Loading rate, tons per acre		Nitrogen loading rate		Crop	Reclamation project
		Acres	Ha							dT	dMT	Cake dT	dMT	lbs/Acre	Kg/Ha		
450	N 1/2 SEC 22 T6S R57W	219	88.6	ELBERT	CAKE/MAC	2/21/00	3/25/00	169	3,745	618	560.65	2.57	5.76	73	81.84	WHEAT	N
451	S 1/2 SEC 22 T6S R57W	219	88.6	ELBERT	CAKE	2/21/00	3/26/00	93	2,064	327	296.65	1.49	3.34	41	45.97	WHEAT	N
452	N 1/2 SEC 23 T6S R57W	303	122.6	ELBERT	CAKE/MAC	12/26/99	1/13/00	180	4,014	648	587.87	1.95	4.37	56	62.78	WHEAT	N
453	S 1/2 SEC 23 T6S R57W	303	122.6	ELBERT	CAKE/MAC	1/7/00	1/19/00	133	2,980	502	455.41	1.43	3.21	42	47.09	WHEAT	N
454	N 1/2 SEC 24 T6S R57W	30	12.1	ELBERT	CAKE	1/10/00	1/11/00	19	415	64	58.06	2.13	4.78	57	63.90	WHEAT	N
455	S 1/2 SEC 24 T6S R57W	35	14.2	ELBERT	CAKE	1/19/00	1/21/00	28	630	94	85.28	2.69	6.03	74	82.96	WHEAT	N
458	N 1/2 SEC 30 T6S R57W	106.2	43.0	ELBERT	CAKE/MAC	1/5/99	9/4/01	317	7,058	1,391	1,261.92	13.10	29.37	--	--	--	--
461	S 1/2 SEC 29 T6S R57W	23.9	9.7	ELBERT	CAKE/MAC	1/9/99	8/24/01	76	1,696	294	266.72	12.30	27.58	--	--	--	--
462	N 1/2 SEC 28 T6S R57W	107	43.3	ELBERT	CAKE/MAC	1/23/00	2/6/00	89	2,003	346	313.89	2.73	6.12	87	97.54	WHEAT	N
463	S 1/2 SEC 28 T6S R57W	54	21.9	ELBERT	CAKE/MAC	1/26/00	1/29/00	43	923	152	137.89	2.56	5.74	73	81.84	WHEAT	N
464	N 1/2 SEC 27 T6S R57W	213	86.2	ELBERT	CAKE/MAC	1/29/00	2/29/00	169	3,737	605	548.86	2.77	6.21	78	87.45	WHEAT	N
465	S 1/2 SEC 27 T6S R57W	172	69.6	ELBERT	CAKE/MAC	1/21/00	1/26/00	138	3,023	466	422.76	2.56	5.74	72	80.72	WHEAT	N
474	N 1/2 SEC 33 T6S R57W	52.2	21.1	ELBERT	CAKE/MAC	2/7/00	2/9/00	31	697	119	107.96	1.99	4.46	58	65.02	WHEAT	N
476	N 1/2 SEC 34 T6S R57W	61.6	24.9	ELBERT	CAKE/MAC	2/4/00	2/8/00	50	1,129	181	164.20	2.69	6.03	83	93.05	WHEAT	N

Table 2. Methods used to analyze biosolids and soil samples collected near Deer Trail, Colorado, 2000

Element or property	Medium	Analytical method	Reference
Arsenic	Soils and biosolids	HG-AAS ¹	Hageman and Welsch (1996)
Cadmium	Biosolids	ICP-MS ²	Briggs and Meier (1999)
Cadmium	Soils	ICP-AES ³	Motooka (1996)
Copper	Biosolids	ICP-MS ²	Briggs and Meier (1999)
Copper	Soils	ICP-AES ³	Briggs (1996)
Lead	Biosolids	ICP-MS ²	Briggs and Meier (1999)
Lead	Soils	ICP-AES ³	Briggs (1996)
Mercury	Soils and biosolids	CV-AAS ⁴	O'Leary and others (1996)
Molybdenum	Biosolids	ICP-MS ²	Briggs and Meier (1999)
Molybdenum	Soils	ICP-AES ³	Motooka (1996)
Nickel	Biosolids	ICP-MS ²	Briggs and Meier (1999)
Nickel	Soils	ICP-AES ³	Briggs (1996)
Selenium	Soils and biosolids	HG-AAS ¹	Hageman and Welsch (1996)
Zinc	Biosolids	ICP-MS ²	Briggs and Meier (1999)
Zinc	Soils	ICP-AES ³	Briggs (1996)
Gross Alpha, Total	Soils and biosolids	Radiological method	Greenberg (1992)
Gross Beta, Total	Soils and biosolids	Radiological method	Greenberg (1992)
Plutonium-238, Total	Soils and biosolids	Radiological method	Whittaker and Grothaus (1979); Lyon (1980)
Plutonium-239+240, Total	Soils and biosolids	Radiological method	Whittaker and Grothaus (1979); Lyon (1980)

¹Hydride Generation-Atomic Absorption Spectrometry.²Inductively Coupled Plasma-Mass Spectrometry.³Inductively Coupled Plasma-Atomic Emission Spectrometry.⁴Continuous Flow-Cold Vapor-Atomic Absorption Spectrometry.

Table 3. Chemical and radioactivity data for biosolids samples collected at the Metro Wastewater Reclamation District during 2000

[ppm, parts per million; pCi/g, picocuries per gram; ±, plus or minus the analytical uncertainty; maximum allowable concentrations from Colorado Department of Public Health and Environment, 1998]

Element or property	Maximum allowable for Grade I	March	June	August	September	October	November	December
Arsenic, ppm	41	1.7	1.8	2.0	2.0	1.8	1.8	1.7
Cadmium, ppm	39	3.0	2.8	2.7	2.9	2.9	5.4	3.6
Copper, ppm	1,500	530	560	580	560	570	570	610
Lead, ppm	300	64	66	78	75	120	70	65
Mercury, ppm	17	1.6	1.6	2.7	1.5	1.6	1.7	1.6
Molybdenum, ppm	No standard set for Grade I, 75 for Grade II	22	26	31	33	31	23	20
Nickel, ppm	420	30	31	33	30	29	27	28
Selenium, ppm	100	9.8	11	12	11	12	11	10
Zinc, ppm	2,800	610	630	700	690	670	690	700
Gross Alpha, pCi/g	40	23 ± 14	44 ± 11	36 ± 12	45 ± 12	43 ± 11	44 ± 11	36 ± 11
Gross Beta, pCi/g	No standard set for Grade I	26 ± 5	23 ± 6	27 ± 5	29 ± 5	29 ± 5	25 ± 4	22 ± 4
Plutonium 238, pCi/g	No standard set for Grade I	0.00 ± 0.01	0.00 ± 0.01	0.02 ± 0.03	0.01 ± 0.02	0.00 ± 0.01	-0.01 ± 0.01	0.01 ± 0.02
Plutonium 239+240, pCi/g	No standard set for Grade I	0.00 ± 0.01	0.00 ± 0.01	0.00 ± 0.01	0.00 ± 0.01	-0.01 ± 0.01	0.00 ± 0.01	0.00 ± 0.01

Table 4. Comparison of radioactivity data from two laboratories for biosolids samples

[Samples were run at two different laboratories for quality-assurance purposes; concentrations in picocuries per gram]

Constituent or property	Data from Acculabs received in 2001 (reported in table 3)	Data from Severn Trent Laboratory received in 2002 ¹
June 2000 biosolids sample		
Gross alpha	44 ± 11	40.2 ± 10
Gross beta	23 ± 6	30.1 ± 5.2
Plutonium-238	0.00 ± 0.01	0.00 ± 0.02
Plutonium-239+240	0.00 ± 0.01	0.00 ± 0.00
NIST Standard Reference Material 2781		
Gross alpha	34 ± 13, 40 ± 11, 51 ± 13, 37 ± 11, 60 ± 13, 44 ± 12, 42 ± 12	34 ± 9, 39 ± 10, 45 ± 11
Gross beta	28 ± 7, 27 ± 6, 35 ± 6, 31 ± 5, 28 ± 5, 30 ± 5, 29 ± 5	21 ± 4, 23 ± 4, 23 ± 4
Plutonium-238	0.00 ± 0.01, 0.00 ± 0.00, 0.00 ± 0.01, 0.00 ± 0.01, 0.01 ± 0.02, 0.00 ± 0.01, 0.01 ± 0.02	-0.001 ± 0.002, 0.00 ± 0.025, 0.00 ± 0.024
Plutonium-239+240	0.01 ± 0.02, 0.00 ± 0.01, 0.00 ± 0.01, 0.00 ± 0.01, 0.00 ± 0.01, 0.00 ± 0.01, 0.00 ± 0.00	0.001 ± 0.002, 0.00 ± 0.027, 0.01 ± 0.02

¹ U.S. Geological Survey contract laboratory used for all U.S. Geological Survey water resources radiochemical analyses in 1999 through 2002.**Table 5.** Trace-element data for wheat whole-plant samples collected from the soil-monitoring sites near Deer Trail, Colorado, September 2000

[All data in milligrams per kilogram; <, less than]

Element	Arapahoe County site			Elbert County site		
	North (control) field	Middle (biosolids application) field	South (control) field	North (control) field	Middle (biosolids application) field	South (control) field
Arsenic	0.12	0.15	0.28	<0.05	<0.05	<0.05
Cadmium	.05	.13	.14	.03	.03	.02
Copper	5.4	5.8	5.4	5.5	4.0	4.0
Lead	.17	.30	.37	.06	.04	.07
Mercury	.006	.005	.008	.006	.007	.006
Molybdenum	.86	.86	.79	1.71	1.06	1.08
Nickel	.94	1.20	1.25	.51	.45	.44
Selenium	.62	.44	.36	1.70	.40	.16
Zinc	17	15	15	21	15	16

44 Data for a Biosolids-Application Area near Deer Trail, Colorado, 2000

Table 6. Trace-element and radioactivity data for wheat-grain samples collected from the soil-monitoring sites near Deer Trail, Colorado, September 2000

[mg/kg, milligram per kilogram; pCi/g, picocuries per gram; <, less than; $\pm$, plus or minus the analytical uncertainty]

Element or property	Arapahoe County site		Elbert County site		
	South (control) field	Middle (biosolids application) field	North (control) field	Middle (biosolids application) field	South (control) field
Arsenic, mg/kg	<0.05	<0.05	<0.05	<0.05	<0.05
Cadmium, mg/kg	.03	.04	.02	.02	.02
Copper, mg/kg	5.1	6.6	4.9	4.8	4.5
Lead, mg/kg	.01	.02	<.01	.01	<.01
Mercury, mg/kg	<.004	<.004	<.004	<.004	<.004
Molybdenum, mg/kg	.43	.80	.70	.51	.61
Nickel, mg/kg	.74	1.35	.43	.50	.50
Selenium, mg/kg	.41	1.4	2.1	.64	.38
Zinc, mg/kg	20	20	26	25	22
Gross Alpha, pCi/g	--	--	1 $\pm$ -1	1 $\pm$ -1	1 $\pm$ -1
Gross Beta, pCi/g	--	--	6 $\pm$ -2	6 $\pm$ -2	5 $\pm$ -2
Plutonium-238, pCi/g	--	--	0.00 $\pm$ -0.03	0.00 $\pm$ -0.01	0.00 $\pm$ -0.01
Plutonium-239+240, pCi/g	--	--	0.00 $\pm$ -0.00	0.00 $\pm$ -0.00	0.00 $\pm$ -0.01

Table 7. Trace-element data for millet-grain samples collected from the Arapahoe County soil-monitoring site near Deer Trail, Colorado, September 2000

[in milligrams per kilogram; <, less than]

Element	Arapahoe County site		
	North (control) field	Middle (biosolids application) field	South (control) field
Arsenic	0.07	0.04	0.06
Cadmium	.05	.05	.06
Copper	7.0	7.1	7.4
Lead	.21	.02	.06
Mercury	<.004	<.004	<.004
Molybdenum	.73	.78	.65
Nickel	6.4	6.5	5.3
Selenium	.26	.24	.27
Zinc	22	19	21

Table 8. Methods used to analyze ground-water samples collected near Deer Trail, Colorado, 2000

[MRL, minimum reporting level, and dilutions for samples having high specific conductance may result in higher MRL's for some samples; MDC, minimum detectable concentration (radiochemical samples); ICP, inductively coupled plasma; AA, atomic absorption spectrometry; MS, mass spectroscopy; ASF, automated segmented-flow spectrophotometry; IC, ion chromatography; $\mu\text{S}/\text{cm}$, microsiemens per centimeter at 25 degrees Celsius; mg/L , milligrams per liter; $\mu\text{g}/\text{L}$, micrograms per liter; GF, graphite furnace; pCi/L , picocuries per liter]

Property or constituent	Units	Analytical method	MRL or MDC	Method Reference
Major ions and mineral characteristics				
Specific conductance, lab	$\mu\text{S}/\text{cm}$	Wheatstone bridge	2.6	Fishman and Friedman (1989)
pH, lab	units	Electrometric electrode	0.1	Fishman and Friedman (1989)
Calcium, dissolved	mg/L	ICP	0.02	Fishman (1993)
Magnesium, dissolved	mg/L	ICP	0.014	Fishman (1993)
Sodium, dissolved	mg/L	ICP	0.09	Fishman (1993)
Potassium, dissolved	mg/L	AA	0.24	Fishman and Friedman (1989)
Acid-neutralizing capacity, lab as CaCO_3	mg/L	Electrometric titration	1	Fishman and Friedman (1989)
Sulfate, dissolved	mg/L	IC	0.31	Fishman and Friedman (1989)
Chloride, dissolved	mg/L	IC	0.29	Fishman and Friedman (1989)
Fluoride, dissolved	mg/L	Colorimetry, ASF, ion-selective electrode	0.1	Fishman and Friedman (1989)
Bromide, dissolved	mg/L	Colorimetry, ASF, fluorescein	0.01	Fishman and Friedman (1989)
Silica, dissolved	mg/L	Colorimetry, ASF, molybdate blue	0.09	Fishman (1993)
Dissolved solids, residue at 180°C	mg/L	Gravimetric	10	Fishman and Friedman (1989)
Nutrients				
Nitrite plus nitrate, dissolved as N	mg/L	Colorimetry, ASF, cadmium reduction, diazotization	0.037	Fishman (1993)
Nitrogen, ammonia, dissolved as N	mg/L	Colorimetry, ASF, salicylate-hypochlorite	0.029	Fishman (1993)
Nitrogen, ammonia plus organic, total as N	mg/L	Colorimetry, ASF, microkjeldahl digestion	0.1	Unpublished
Nitrogen, ammonia plus organic, dissolved as N	mg/L	Colorimetry, ASF, microkjeldahl digestion	0.1	Patton and Truitt (1992)
Phosphorus, total as P	mg/L	Colorimetry, ASF, microkjeldahl digestion	0.05	Patton and Truitt (1992)
Phosphorus, dissolved as P	mg/L	Colorimetry, ASF, microkjeldahl digestion	0.05	Patton and Truitt (1992)
Trace elements				
Aluminum, dissolved as Al	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)
Antimony, dissolved as Sb	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)
Arsenic, dissolved as As	$\mu\text{g}/\text{L}$	Hydride generation	2	Jones and Garbarino (1999)
Barium, dissolved as Ba	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)
Beryllium, dissolved as Be	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)

46 Data for a Biosolids-Application Area near Deer Trail, Colorado, 2000**Table 8.** Methods used to analyze ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[MRL, minimum reporting level, and dilutions for samples having high specific conductance may result in higher MRL's for some samples; MDC, minimum detectable concentration (radiochemical samples); ICP, inductively coupled plasma; AA, atomic absorption spectrometry; MS, mass spectroscopy; ASF, automated segmented-flow spectrophotometry; IC, ion chromatography; $\mu\text{S}/\text{cm}$, microsiemens per centimeter at 25 degrees Celsius; mg/L , milligrams per liter; $\mu\text{g}/\text{L}$, micrograms per liter; GF, graphite furnace; pCi/L , picocuries per liter]

Property or constituent	Units	Analytical method	MRL or MDC	Method Reference
Boron, dissolved as B	$\mu\text{g}/\text{L}$	ICP-MS	16	Struzeski, DeGiacomo, and Zayhowski (1996)
Cadmium, dissolved as Cd	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)
Chromium, dissolved as Cr	$\mu\text{g}/\text{L}$	GF AA	0.8	McLain (1993)
Cobalt, dissolved as Co	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)
Copper, dissolved as Cu	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)
Iron, dissolved as Fe	$\mu\text{g}/\text{L}$	ICP	3	Fishman (1993)
Lead, dissolved as Pb	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)
Manganese, dissolved as Mn	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)
Mercury, dissolved as Hg	$\mu\text{g}/\text{L}$	Hydride generation	0.2	Fishman and Friedman (1989)
Molybdenum, dissolved as Mo	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)
Nickel, dissolved as Ni	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)
Selenium, dissolved as Se	$\mu\text{g}/\text{L}$	Hydride generation	2.4	Jones and Garbarino (1999)
Silver, dissolved as Ag	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)
Strontium, dissolved as Sr	$\mu\text{g}/\text{L}$	ICP-MS	1	Fishman (1993)
Zinc, dissolved as Zn	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)
Radioactivity				
Uranium, natural dissolved	$\mu\text{g}/\text{L}$	ICP-MS	1	Faires (1993)
Plutonium-238, total	pCi/L	Alpha spectrometry	0.1*	Not Available
Plutonium-239+240, total	pCi/L	Alpha spectrometry	0.1*	Not Available

*Contract required MDC; reported value may be lower depending upon the sample.

Table 9. Lithologic description for well DTX11 in 2000 near Deer Trail, Colorado

[Description from geologist’s notes; cm, centimeter; bottom 6 feet of borehole sloughed with wet, sandy silt before well could be completed]

Depth below land surface	Source of informa- tion	Description of material
0-20 feet	Core	Medium-brown and dark-brown, dry, silty clay grading into damp, medium-brown, sandy silt.
20-23 feet	Core	Wet, brown silt and very fine-grained sand.
23-24.3 feet	Core	Dry, hard, orange clay grading into very dark gray clay; some small shale pebbles and sandstone fragments.
24.3-28 feet	Core	Very wet, granitic, interbedded coarse sand with pebbles and wet, gray, fine-grained sand and silt.
28-38 feet	Core	Wet, mostly gray, fine-grained sand and silt with a few thin (less than 1 inch) beds of coarse granitic sand and pebbles.
38-40 feet	Core	Dry, crumbly, gray and black striped bedrock (very fine-grained sandy silt interbedded with shale) grading into dry, crumbly, black shale.

Table 10. Information for U.S. Geological Survey monitoring wells near Deer Trail, Colorado, 2000

[Latitude and longitude are in the format degrees minutes seconds referenced to NAD 83; bmp, below measuring point (stickup); Wells shown in bold are sampled for water quality; Elev., elevation in feet above NAVD 88; Metro, Metro Wastewater Reclamation District]

Well	Latitude	Longitude	County	Date drilled	Stickup (feet)	Total depth (feet bmp)	Screened Interval			Screen length (feet)	Screen opening (inches)	Sump length (feet)	Elev. of measuring point (feet)	Property owner	Well diam-eter (inches)
							Top (feet bmp)	Bottom (feet bmp)							
DTX1	394333	1035251	ARAPAHOE	02/16/99	2.56	25.50	20.59	22.19	1.60	0.010	3.31	4,909	Metro	2	
DTX2*	394149	1035138	ARAPAHOE	02/16/99	3.23	20.50	15.59	17.19	1.60	0.010	3.31	4,903	Metro	2	
DTX3	393024	1034328	ELBERT	02/12/99	3.11	18.71	13.80	15.40	1.60	0.010	3.31	5,195	Metro	2	
DTX4	393358	1034342	ELBERT	02/10/99	2.70	16.72	11.81	13.41	1.60	0.010	3.31	4,957	Metro	2	
DTX5*	393358	1034548	ELBERT	02/10/99	2.30	20.90	16.09	17.69	1.60	0.010	3.21	4,975	Metro	2	
DTX6	393358	1034648	ELBERT	02/9/99	2.43	39.21	34.30	35.90	1.60	0.010	3.31	4,970	Metro	2	
DTX7	394054	1035646	ARAPAHOE	02/18/99	2.77	16.10	11.19	12.79	1.60	0.010	3.31	5,075	Price	2	
DTX8A	394054	1035645	ARAPAHOE	03/2/99	2.46	77.52	67.56	71.83	4.27	0.010	5.69	5,076	Price	2	
DTX8B	394054	1035645	ARAPAHOE	03/2/99	2.49	177.48	167.52	171.79	4.27	0.010	5.69	5,076	Price	2	
DTX9*	393902	1035540	ARAPAHOE	02/17/99	2.46	30.15	22.72	24.32	1.60	0.010	5.83	5,121	Weisensee	2	
DTX10A*	393902	1035540	ARAPAHOE	03/4/99	2.03	61.97	52.01	56.28	4.27	0.010	5.69	5,122	Weisensee	2	
DTX10B	393902	1035540	ARAPAHOE	03/4/99	2.11	121.73	111.77	116.04	4.27	0.010	5.69	5,122	Weisensee	2	
DTX11*	393902	1035540	ARAPAHOE	01/19/00	2.24	32	28	30	1.85	0.020	2.35	5,122	Weisensee	2	
D6	393633	1035122	ARAPAHOE	09/12/93	2.65	25	15	25	10	0.010	0.3	5,128.78	Metro	2	
D11a	393345	1035423	ELBERT	10/23/97	2.46	143.03	112.65	122.65	10	0.010	20.38	5,377	Metro	2	
D13	393442	1035438	ARAPAHOE	04/4/94	1.81	16	6	16	10	0.010	0.3	5,235.33	Metro	2	
D17	393334	1035436	ELBERT	04/5/94	0.90	21	11	21	10	0.010	0.3	5,277.73	Metro	2	
D25*	393702	1035442	ARAPAHOE	05/1/95	2.23	23	13	23	10	0.010	0.3	5,167.13	Metro	2	
D29	393641	1035248	ARAPAHOE	11/4/97	2.38	183.19	147.81	157.81	10	0.010	25.38	5,371	Metro	2	
D30	393655	1035122	ARAPAHOE	05/5/95	1.98	19	9	19	10	0.010	0.3	5,096.43	Metro	2	

*Wells having continuous-recorder equipment.

Table 11. Monthly water-level data for U.S. Geological Survey monitoring wells near Deer Trail, Colorado, 2000

[Et1, Solinst electric tape; Et2, Slope Indicator electric tape; St, Steel tape; --, no data; *, continuous-recorder site; W.L. bmp, water level below measuring point, in feet]

Well no.	January (01/12/00)	February (02/04/00)	March (03/03/00)	April (04/05/00)	May (05/08/00)	June (06/02/00)	July (07/06/00)	August (08/03/00)	August (08/03/00)	August (08/03/00)	September (09/01/00)	October (10/10/00- 10/18/00)	November (11/03/00)	December (12/02/00)
	W.L. bmp, Et1	W.L. bmp, Et1	W.L. bmp, Et1	W.L. bmp, Et1	W.L. bmp, Et1	W.L. bmp, Et1	W.L. bmp, Et1	W.L. bmp, Et1	W.L. bmp, Et2	W.L. bmp, St	W.L. bmp, Et1	W.L. bmp, Et1	W.L. bmp, Et1	W.L. bmp, Et1
DTX1	7.87	7.84	7.84	7.53	7.65	7.99	8.90	9.03	--	--	9.23	9.80	9.69	9.60
DTX2*	7.25	7.31	7.24	7.07	7.39	8.00	10.37	8.13	--	--	8.76	8.86	8.47	8.40
DTX3	8.90	9.01	9.11	9.05	9.29	9.42	9.62	9.86	9.81	9.83	10.04	10.24	9.82	10.31
DTX4	7.94	8.09	8.05	7.77	7.98	9.69	9.89	11.01	11.02	11.01	11.98	12.51	12.62	12.69
DTX5*	9.74	9.90	9.88	9.45	9.73	10.15	10.26	11.19	--	--	11.72	11.97	12.00	12.04
DTX6	21.67	21.75	21.74	21.45	21.39	21.56	21.79	21.95	--	--	22.20	22.49	22.57	22.61
DTX7	7.56	7.52	7.44	7.25	7.31	7.66	8.57	8.45	--	--	8.71	8.68	8.47	8.30
DTX8A	7.92	7.88	7.75	7.48	7.77	8.42	9.35	9.15	--	--	9.58	9.58	9.33	8.73
DTX8B	4.78	4.88	4.82	4.63	4.70	4.83	4.88	4.86	--	--	4.89	4.97	4.82	4.81
DTX9	12.81	12.87	12.85	12.81	12.88	12.96	13.00	12.77	12.74	12.74	12.76	12.89	12.87	12.84
DTX10A	13.00	13.09	13.07	12.97	13.11	13.23	13.31	12.93	--	--	13.00	13.12	13.09	13.07
DTX10B	18.67	18.71	18.60	18.32	18.40	18.49	18.47	18.27	--	--	18.54	18.74	18.45	18.40
DTX11*	--	--	13.41	13.37	13.20	13.29	13.35	13.11	--	--	13.10	13.23	13.20	13.17
D6	9.54	9.61	9.61	9.51	9.18	9.20	9.42	9.31	9.29	--	9.41	9.54	9.48	9.24
D11a	113.24	113.25	113.25	113.00	113.28	113.30	113.16	113.13	--	--	113.10	112.82	113.21	113.11
D13	7.46	7.75	7.16	6.63	6.92	7.37	8.08	6.82	6.81	6.80	8.61	8.47	8.19	7.68
D17	10.99	11.00	10.99	10.91	10.74	10.77	11.08	11.52	11.51	11.44	12.41	12.65	12.60	12.48
D25*	10.35	10.37	10.37	10.24	10.19	10.29	10.68	7.13	--	--	7.98	8.81	9.06	9.27
D29	154.25	154.74	154.52	154.24	154.32	154.67	154.35	154.38	--	--	154.29	154.19	154.49	154.57
D30	5.19	5.20	5.10	4.64	4.87	5.26	5.98	4.95	4.95	4.94	5.28	5.38	4.52	4.64

50 Data for a Biosolids-Application Area near Deer Trail, Colorado, 2000
Table 12. Field-parameter data associated with ground-water sampling near Deer Trail, Colorado, 2000

[$\mu\text{S}/\text{cm}$, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter, --, no data because equipment malfunctioned]

Well	Sample date (mm/dd/yy)	Sample time (hhmm)	Specific conduc- tance, field ($\mu\text{S}/\text{cm}$)	pH, field (stan- dard units)	Water temperature (degrees Celsius)	Water level before sampling, depth below measuring point (feet)	Oxygen, dissolved (mg/L)
DTX1	01/10/00	1345	4,200	7.0	11.1	7.84	0.50
DTX1	04/14/00	1220	4,210	7.0	11.3	7.46	0.52
DTX1	07/07/00	940	4,310	6.7	11.1	8.92	0.62
DTX1	10/17/00	1200	4,060	7.1	12.3	9.80	0.41
DTX2	01/10/00	1520	4,300	6.9	10.1	7.24	0.90
DTX2	04/14/00	1345	4,320	6.9	11.0	7.16	0.76
DTX2	07/07/00	1050	4,510	6.6	12.1	9.16	0.88
DTX2	10/17/00	1005	4,210	6.9	12.5	8.86	0.90
DTX3	01/10/00	1035	1,450	7.0	10.8	8.85	2.7
DTX3	04/11/00	1015	1,840	6.9	9.8	9.17	3.24
DTX3	07/10/00	1030	2,130	6.6	12.4	9.70	3.56
DTX3	10/11/00	1030	2,050	7.0	14.6	10.24	3.24
DTX4	01/06/00	955	3,000	6.6	10.1	7.93	0.90
DTX4	04/11/00	1145	3,070	6.7	8.7	7.25	1.07
DTX4	07/11/00	1010	3,210	6.4	11.6	10.12	1.33
DTX4	10/11/00	1215	3,180	6.7	14.5	12.51	0.96
DTX5	01/06/00	1310	2,950	6.8	10.8	9.78	0.90
DTX5	04/11/00	1345	2,820	6.9	10.3	9.65	0.90
DTX5	07/11/00	1150	3,100	6.6	12.5	10.80	1.23
DTX5	10/16/00	1015	2,950	6.8	12.6	11.97	1.07
DTX6	01/06/00	1530	4,150	6.9	11.7	21.66	0.80
DTX6	04/14/00	1000	4,380	7.0	12.7	21.11	1.01
DTX6	07/13/00	1300	4,500	6.7	13.5	21.90	1.35
DTX6	10/11/00	1400	4,290	7.0	13.4	22.49	1.00
DTX8A	01/13/00	1315	1,850	7.4	11.4	7.97	0.40
DTX8A	04/13/00	1415	1,850	7.3	15.4	7.54	--
DTX8A	07/10/00	1505	1,950	7.2	18.8	9.47	0.62
DTX8A	10/16/00	1435	1,830	7.3	15.1	9.58	0.95

Table 12. Field-parameter data associated with ground-water sampling near Deer Trail, Colorado, 2000—Continued

[μS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter, --, no data because equipment malfunctioned]

Well	Sample date (mm/dd/yy)	Sample time (hhmm)	Specific conduct- tance, field (μS/cm)	pH, field (stan- dard units)	Water temperature (degrees Celsius)	Water level before sampling, depth below measuring point (feet)	Oxygen, dissolved (mg/L)
DTX10A	01/11/00	1130	3,150	7.0	13.2	12.95	0.20
DTX10A	04/17/00	1415	3,120	6.9	17.0	13.03	0.42
DTX10A	07/13/00	1020	3,220	6.8	20.2	13.42	0.26
DTX10A	10/18/00	1040	3,070	7.1	16.2	13.12	3.04
D6	01/11/00	1355	16,000	7.0	12.0	9.35	0.30
D6	04/13/00	945	16,600	6.9	11.7	9.31	0.77
D6	07/11/00	1345	17,500	6.7	13.4	9.44	0.98
D6	10/12/00	1300	16,100	6.9	12.7	9.54	0.70
D13	01/12/00	1005	1,460	7.0	8.5	7.46	0.50
D13	04/17/00	1115	1,390	7.0	9.0	6.59	0.44
D13	07/07/00	1230	1,500	6.8	13.2	8.13	0.76
D13	10/16/00	1215	1,670	7.1	13.8	8.47	1.52
D17	01/07/00	1500	470	7.5	12.1	10.99	1.50
D17	04/10/00	1415	480	7.4	12.3	10.84	2.46
D17	07/06/00	1320	540	7.3	14.1	11.11	1.74
D17	10/10/00	1245	480	7.6	14.4	12.65	0.49
D25	01/12/00	1245	4,300	7.0	11.8	10.32	0.40
D25	04/10/00	1615	4,260	6.9	11.3	10.23	0.66
D25	07/06/00	1500	4,620	6.8	13.2	10.69	0.71
D25	10/10/00	1455	4,910	7.1	13.8	8.81	0.39
D29	01/07/00	1130	4,020	6.7	17.6	154.41	3.0
D29	04/10/00	1215	3,940	6.5	20.0	154.35	3.51
D29	07/06/00	1105	4,090	6.5	23.1	154.35	1.74
D29	10/12/00	1110	3,960	6.6	20.8	154.19	1.08
D30	01/11/00	1600	4,870	6.9	10.7	5.10	0.30
D30	04/13/00	1200	5,060	6.8	10.2	4.75	0.49
D30	07/10/00	1245	5,110	6.7	12.1	6.07	0.59
D30	10/12/00	1450	4,960	7.0	14.4	5.38	0.40

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000

[$\mu\text{S}/\text{cm}$, microsiemens per centimeter at 25 degrees Celsius; mg/L , milligrams per liter; $\mu\text{g}/\text{L}$, micrograms per liter; $\text{p}^{\text{C}}\text{V}_L$, picocuries per liter; $\leq$, less than; ND, no data available; E, value estimated by laboratory; -, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Specific conductance, lab ($\mu\text{S}/\text{cm}$)	pH, laboratory (standard units)	Hardness total (mg/L as CaCO_3)	Calcium, dissolved (mg/L as Ca)	Magnesium, dissolved (mg/L as Mg)	Potassium, dissolved (mg/L as K)	Sodium adsorption ratio	Sodium, dissolved (mg/L as Na)	Sodium (percent)	Acid-neutralizing capacity, titration to pH 4.5, lab (mg/L as CaCO_3)	Chloride, dissolved (mg/L as Cl)
DTX1	01/10/00	1345	4,100	7.3	2,100	477	221	3.5	4	372	28	319	49.7
DTX1	04/14/00	1220	4,150	7.2	2,000	453	198	3.3	3	331	27	318	47.5
DTX1	07/07/00	0940	4,050	7.3	2,000	459	209	3.1	3	355	28	315	49.9
DTX1	10/17/00	1200	4,070	7.3	1,700	467	124	3.5	4	331	30	316	49.1
DTX2	01/10/00	1520	4,200	7.1	2,000	460	194	8.6	4	364	29	406	45.6
DTX2	04/14/00	1345	4,240	7.0	2,000	495	194	7.3	3	360	28	404	40.2
DTX2	07/07/00	1050	4,250	7.1	2,100	472	214	7.8	4	392	29	405	36.0
DTX2	10/17/00	1005	4,230	7.0	1,700	479	122	8.5	4	382	33	409	40.2
DTX3	01/10/00	1035	1,410	7.4	720	170	69.9	6.2	.9	58.2	15	274	17.1
DTX3	04/11/00	1015	1,880	7.3	990	240	94.2	7.3	1	72.8	14	314	26.0
DTX3	07/10/00	1030	2,020	7.4	1,100	266	102	7.3	1	94.1	16	328	28.6
DTX3	10/11/00	1030	2,100	7.1	1,100	263	102	7.5	1	99.5	17	338	28.7
DTX4	01/06/00	0955	2,950	7.0	1,700	531	86.9	8.8	2	160	17	353	15.2
DTX4	04/11/00	1145	3,140	6.9	1,800	551	90.8	7.2	2	158	16	424	16.1
DTX4	07/11/00	1010	3,070	6.9	1,700	526	89.1	7.5	2	169	18	346	17.4
DTX4	10/11/00	1215	3,200	6.9	1,800	563	95.4	8.8	2	179	18	348	16.1

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[$\mu\text{S}/\text{cm}$, microsiemens per centimeter at 25 degrees Celsius; mg/L , milligrams per liter; $\mu\text{g}/\text{L}$, micrograms per liter; pCi/L , picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hhmm)	Specific conductance, lab ($\mu\text{S}/\text{cm}$)	pH, laboratory (standard units)	Hardness total (mg/L as CaCO_3)	Calcium, dissolved (mg/L as Ca)	Magnesium, dissolved (mg/L as Mg)	Potassium, dissolved (mg/L as K)	Sodium adsorption ratio	Sodium, dissolved (mg/L as Na)	Sodium (percent)	Acid-neutralizing capacity, titration to pH 4.5, lab (mg/L as CaCO_3)	Chloride, dissolved (mg/L as Cl)
DTX5	01/06/00	1310	2,900	7.1	1,800	566	86.2	4.6	1	94.6	10	275	8.6
DTX5	04/11/00	1345	2,860	7.1	1,800	568	81.8	4.9	.9	91.0	10	270	7.7
DTX5	07/11/00	1150	2,970	7.1	1,700	559	84.2	4.6	1	120	13	273	10.6
DTX5	10/16/00	1015	2,900	7.1	1,600	523	81.1	5.1	2	148	16	275	10.0
DTX6	01/06/00	1530	4,100	7.2	2,200	468	243	12.8	3	307	23	270	19.2
DTX6	04/14/00	1000	4,300	7.1	2,200	476	248	13.6	3	309	23	274	21.8
DTX6	07/13/00	1300	4,300	7.3	2,300	468	265	12.6	3	336	24	270	20.6
DTX6	10/11/00	1400	4,290	7.1	2,300	480	263	12.5	3	328	24	270	20.3
DTX8A	01/13/00	1315	1,860	7.6	490	144	32.0	6.6	5	240	51	225	26.9
DTX8A	04/13/00	1415	1,920	7.6	460	135	29.9	7.0	4	219	50	224	27.2
DTX8A	07/10/00	1505	1,870	7.8	510	149	32.4	6.1	5	241	50	216	26.3
DTX8A	10/16/00	1435	1,890	7.5	490	144	31.6	6.2	5	232	50	224	25.6
DTX10A	01/11/00	1130	3,070	7.0	1,800	443	170	¹ 12.5	2	150	15	226	17.4
DTX10A	04/17/00	1415	3,180	7.2	1,800	446	171	8.1	2	147	15	223	16.3
DTX10A	07/13/00	1020	3,140	7.2	1,800	433	170	8.8	2	152	16	222	17.9
DTX10A	10/18/00	1040	3,100	7.2	1,800	456	169	8.3	2	153	15	221	16.6

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[μS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; μg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hhmm)	Specific conductance, lab (μS/cm)	pH, laboratory (standard units)	Hardness total (mg/L as CaCO ₃)	Calcium, dissolved (mg/L as Ca)	Magnesium, dissolved (mg/L as Mg)	Potassium, dissolved (mg/L as K)	Sodium adsorption ratio	Sodium, dissolved (mg/L as Na)	Sodium (percent)	Acid-neutralizing capacity, titration to pH 4.5, lab (mg/L as CaCO ₃)	Chloride, dissolved (mg/L as Cl)
D6	01/11/00	1355	16,300	7.2	10,000	400	2,200	11.2	8	1,860	29	638	411
D6	04/13/00	0945	16,300	7.2	11,000	446	2,320	12.9	9	2,100	30	640	380
D6	07/11/00	1345	16,000	7.2	10,000	414	2,170	12.0	9	2,070	31	638	404
D6	10/12/00	1300	16,200	7.1	10,000	437	2,190	10.4	9	2,110	31	640	419
D13	01/12/00	1005	1,410	7.4	720	198	54.9	2.4	1	58.8	15	248	3.1
D13	04/17/00	1115	1,390	7.4	660	179	50.6	2.5	1	57.1	16	253	2.9
D13	07/07/00	1230	1,420	7.7	680	188	52.1	2.8	1	62.6	17	255	3.2
D13	10/16/00	1215	1,650	7.2	830	226	63.0	2.8	1	66.7	15	256	3.8
D17	01/07/00	1500	492	7.6	210	55.3	18.2	1.6	.5	16.0	14	209	2.3
D17	04/10/00	1415	523	7.7	210	54.9	18.5	1.6	.5	15.4	13	212	2.8
D17	07/06/00	1320	533	7.8	230	56.3	22.5	1.6	.4	15.1	12	215	3.6
D17	10/10/00	1245	484	7.7	220	59.7	18.0	1.6	.5	15.5	13	202	2.9
D25	01/12/00	1245	4,300	7.3	2,500	642	206	7.2	2	260	19	449	108
D25	04/10/00	1615	4,320	7.2	2,400	646	194	7.1	2	240	18	400	119
D25	07/06/00	1500	4,510	7.3	2,600	638	232	7.3	2	286	20	496	99.8
D25	10/10/00	1455	4,900	7.2	2,900	794	219	8.3	3	334	20	585	81.5

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[µS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hhmm)	Specific conductance, lab (µS/cm)	pH, laboratory (standard units)	Hardness total (mg/L as CaCO ₃)	Calcium, dissolved (mg/L as Ca)	Magnesium, dissolved (mg/L as Mg)	Potassium, dissolved (mg/L as K)	Sodium adsorption ratio	Sodium, dissolved (mg/L as Na)	Sodium (percent)	Acid-neutralizing capacity, titration to pH 4.5, lab (mg/L as CaCO ₃)	Chloride, dissolved (mg/L as Cl)
D29	01/7/00	1130	4,020	6.9	2,800	535	346	11.1	1	143	10	293	11.5
D29	04/10/00	1215	4,060	6.7	2,600	521	316	10.9	1	138	10	290	12.3
D29	07/06/00	1105	3,910	6.8	2,600	529	315	10.3	1	137	10	290	13.0
D29	10/12/00	1110	3,980	6.8	2,700	544	313	9.5	1	139	10	283	10.8
D30	01/11/00	1600	4,800	7.1	2,700	428	400	3.9	3	371	23	364	53.1
D30	04/13/00	1200	4,980	7.2	2,800	425	410	4.0	3	361	22	367	50.0
D30	07/10/00	1245	4,640	7.4	2,600	419	386	3.9	3	369	23	344	47.4
D30	10/12/00	1450	4,870	7.1	2,700	439	389	3.6	3	381	23	346	54.0

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[µS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hhmm)	Fluoride, dissolved (mg/L as F)	Silica, dissolved (mg/L as SiO ₂)	Sulfate, dissolved (mg/L as SO ₄)	Nitrogen ammonia plus organic, dissolved (mg/L as N)	Nitrogen ammonia plus organic, total (mg/L as N)	Nitrite plus nitrate (mg/L as N)	Phos-phorus, dissolved (mg/L as P)	Phos-phorus, total (mg/L as P)	Solids, residue on evaporation at 180°C, dissolved (mg/L)	Dissolved solids, sum of constituents (mg/L)
DTX1	01/10/00	1345	0.8	33.9	2,440	0.22	0.23	2.80	0.073	0.070	4,170	3,810
DTX1	04/14/00	1220	.8	30.5	2,400	.24	.21	1.38	.081	.054	4,090	3,670
DTX1	07/07/00	0940	.8	32.0	2,390	.22	.20	1.29	.080	.080	4,000	3,700
DTX1	10/17/00	1200	.7	32.2	2,400	E.09	.26	1.22	.085	.080	4,050	3,610
DTX2	01/10/00	1520	.5	15.9	2,430	.86	.62	.05	<.05	<.05	4,200	3,780
DTX2	04/14/00	1345	.5	16.1	2,410	.89	.84	<.04	<.05	<.05	4,160	3,780
DTX2	07/07/00	1050	.5	16.6	2,480	.88	.85	<.04	<.05	<.05	4,230	3,870
DTX2	10/17/00	1005	.5	16.5	2,440	1<.10	.98	<.04	<.05	E.04	4,140	3,740
DTX3	01/10/00	1035	.4	15.7	516	.19	.16	2.78	<.05	<.05	1,090	1,030
DTX3	04/11/00	1015	.4	16.6	770	<.10	.17	3.80	<.05	<.05	1,550	1,440
DTX3	07/10/00	1030	.4	10.4	860	.20	.19	1<.04	<.05	<.05	1,680	1,570
DTX3	10/11/00	1030	.4	17.8	901	.20	.20	4.23	<.05	<.06	1,760	1,640
DTX4	01/06/00	0955	.2	11.6	1,590	.26	.29	.33	<.05	<.05	2,800	2,620
DTX4	04/11/00	1145	.2	11.5	1,670	<.10	.31	.15	<.05	<.05	3,000	2,760
DTX4	07/11/00	1010	.2	10.7	1,670	.29	.29	4.66	<.05	<.05	2,970	2,720
DTX4	10/11/00	1215	.3	11.6	1,760	.40	.38	6.52	<.05	<.06	3,060	2,880

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[µS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hhmm)	Fluoride, dissolved (mg/L as F)	Silica, dissolved (mg/L as SiO ₂)	Sulfate, dissolved (mg/L as SO ₄)	Nitrogen ammonia plus organic, dissolved (mg/L as N)	Nitrogen ammonia plus organic, total (mg/L as N)	Nitrite plus nitrate (mg/L as N)	Phos-phorus, dissolved (mg/L as P)	Phos-phorus, total (mg/L as P)	Solids, residue on evaporation at 180°C, dissolved (mg/L)	Dissolved solids, sum of constituents (mg/L)
DTX5	01/06/00	1310	0.2	11.7	1,720	0.12	0.16	0.048	<0.05	<0.05	2,920	2,660
DTX5	04/11/00	1345	0.3	11.2	1,670	<0.10	0.18	0.054	<0.05	<0.05	2,790	2,600
DTX5	07/11/00	1150	.3	11.2	1,750	.15	.12	.113	<0.05	<0.05	2,950	2,710
DTX5	10/16/00	1015	.3	11.9	1,640	.15	.18	.116	<0.05	E.04	2,750	2,590
DTX6	01/06/00	1530	0.4	11.4	2,560	0.12	0.12	0.034	<0.05	<0.05	4,210	3,790
DTX6	04/14/00	1000	.5	11.2	2,660	.11	.14	E.015	<0.05	<0.05	4,380	3,910
DTX6	07/13/00	1300	.4	11.3	2,730	.10	.11	.060	<0.05	<0.05	4,440	4,010
DTX6	10/11/00	1400	.5	11.1	2,680	.14	E.10	.087	<0.05	<0.06	4,360	3,960
DTX8A	01/13/00	1315	.4	13.0	767	1.6	1.5	1.41	<0.05	<0.05	1,420	1,370
DTX8A	04/13/00	1415	.4	12.2	748	1.6	1.5	1.41	1.12	<0.05	1,400	1,320
DTX8A	07/10/00	1505	.4	13.5	757	1.5	1.4	1.39	<0.05	<0.05	1,390	1,360
DTX8A	10/16/00	1435	.3	12.9	751	1.3	1.5	1.35	<0.05	<0.06	1,410	1,340
DTX10A	01/11/00	1130	0.8	16.6	1,890	1.4	1.4	1.24	<0.05	<0.05	3,100	2,850
DTX10A	04/17/00	1415	.9	16.8	1,880	1.3	1.4	1.27	<0.05	<0.05	3,080	2,830
DTX10A	07/13/00	1020	.9	16.7	1,880	1.4	1.4	1.28	<0.05	<0.05	3,090	2,820
DTX10A	10/18/00	1040	.8	16.8	1,880	1.4	1.2	1.19	<0.05	<0.06	3,070	2,840
D6	01/11/00	1355	.7	20.1	13,000	1.6	1.6	.067	<0.05	E.04	21,000	183,00
D6	04/13/00	0945	.8	21.3	12,900	1.5	1.4	.045	E.04	E.04	20,300	18,600
D6	07/11/00	1345	.8	20.6	13,100	1.5	1.4	<.030	E.03	E.04	21,100	18,700
D6	10/12/00	1300	.8	19.8	13,100	1.1	1.5	.131	<0.05	E.04	21,500	18,700

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[μS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; μg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hhmm)	Fluoride, dissolved (mg/L as F)	Silica, dissolved (mg/L as SiO ₂)	Sulfate, dissolved (mg/L as SO ₄)	Nitrogen ammonia plus organic, dissolved (mg/L as N)	Nitrogen ammonia plus organic, total (mg/L as N)	Nitrite plus nitrate (mg/L as N)	Phos-phorus, dissolved (mg/L as P)	Phos-phorus, total (mg/L as P)	Solids, residue on evaporation at 180°C, dissolved (mg/L)	Dissolved solids, sum of constituents (mg/L)
D13	01/12/00	1005	1.4	12.7	626	.15	.13	.049	<.05	<.05	1,150	1,110
D13	04/17/00	1115	1.4	11.9	545	.14	.22	<.029	<.05	<.05	1,080	1,000
D13	07/07/00	1230	1.4	13.6	563	0.14	0.13	0.040	<.05	<.05	1,110	1,040
D13	10/16/00	1215	1.4	14.1	719	.14	.17	E.027	<.05	<.06	1,340	1,250
D17	01/07/00	1500	1.6	18.2	44.9	E.10	E.10	E.026	.08	.10	298	294
D17	04/10/00	1415	1.7	17.5	42.4	.12	.12	<.029	.09	.09	305	298
D17	07/06/00	1320	1.7	18.8	40.6	.16	.17	<.029	.08	.07	325	315
D17	10/10/00	1245	1.6	19.0	49.3	E.10	E.10	<.049	E.05	.07	287	292
D25	01/12/00	1245	1.0	27.8	2,460	0.69	0.67	0.045	0.12	0.12	4,520	4,000
D25	04/10/00	1615	1.1	26.3	2,420	.62	.72	.070	.10	.10	4,390	3,900
D25	07/06/00	1500	1.0	28.5	2,520	.74	.77	.138	.15	.15	4,640	4,130
D25	10/10/00	1455	1.0	28.8	2,860	.11	.88	.110	.11	.12	5,270	4,710
D29	01/07/00	1130	.5	21.1	2,690	.45	.51	.446	<.05	E.04	4,260	3,950
D29	04/10/00	1215	.6	20.6	2,630	.44	.47	.428	<.05	<.05	4,280	3,830
D29	07/06/00	1105	.5	20.4	2,620	.45	.49	.466	E.04	E.04	4,300	3,830
D29	10/12/00	1110	.5	20.2	2,660	.41	.45	.431	<.05	.06	4,120	3,880
D30	01/11/00	1600	.7	21.8	3,070	.29	.29	.065	E.04	.06	5,110	4,580
D30	04/13/00	1200	.8	20.7	3,070	.29	.27	.047	E.03	E.05	5,070	4,570
D30	07/10/00	1245	.8	21.0	3,060	.28	.29	.134	<.05	E.04	5,000	4,520
D30	10/12/00	1450	.8	21.8	3,100	.30	.34	.120	<.05	E.06	4,980	4,600

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[μS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; μg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; ~, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Aluminum, dissolved (μg/L as Al)	Anti-mony, dissolved (μg/L as Sb)	Arsenic, dissolved (μg/L as As)	Barium, dissolved (μg/L as Ba)	Beryllium, dissolved (μg/L as Be)	Boron, dissolved (μg/L as B)	Bromide, dissolved (mg/L as Br)	Cadmium, dissolved (μg/L as Cd)	Chromium, dissolved (μg/L as Cr)	Cobalt, dissolved (μg/L as Co)	Copper, dissolved (μg/L as Cu)
DTX1	01/10/00	1345	7	<2	E1	9	<2	670	0.67	<2.0	<1.0	<2	8
DTX1	04/14/00	1220	<4	<1	4	8	<1	530	.71	<1.0	<.8	<1	14
DTX1	07/07/00	0940	<1	<1	E2	8	<1	550	.33	<1.0	<.8	1	9
DTX1	10/17/00	1200	2	<2	E2	8	<2	590	.70	<2.0	<.8	2	12
DTX2	01/10/00	1520	18	<2	<2	16	<2	310	.53	<2.0	<1.0	5	6
DTX2	04/14/00	1345	<3	<1	E2	15	<1	290	.40	<1.0	<.8	4	11
DTX2	07/07/00	1050	<1	<1	E1	14	<1	350	.41	<1.0	<.8	5	8
DTX2	10/17/00	1005	<2	<2	E1	14	<2	360	.48	<2.0	<.8	7	11
DTX3	01/10/00	1035	2	<1	<2	23	<1	210	.22	<1.0	E.6	<1	3
DTX3	04/11/00	1015	<6	<1	<2	23	<1	220	.28	<1.0	<.8	<1	3
DTX3	07/10/00	1030	<6	<1	<2	21	<1	230	.29	<1.0	<.8	<1	5
DTX3	10/11/00	1030	<1	<1	<2	21	<1	230	.27	<1	1.7	1	3
DTX4	01/06/00	0955	<2	<2	E1	14	<2	290	.22	<2.0	<.8	<1	5
DTX4	04/11/00	1145	<29	<2	E1	13	<2	270	.23	<2.0	<1.0	<2	6
DTX4	07/11/00	1010	<1	<2	<2	12	<2	240	.29	<2.0	<.8	<2	7
DTX4	10/11/00	1215	<2	E.1	<2	14	<1	310	.31	E.1	2.6	1	6

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[pS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; -, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Aluminum, dissolved (µg/L as Al)	Antimony, dissolved (µg/L as Sb)	Arsenic, dissolved (µg/L as As)	Barium, dissolved (µg/L as Ba)	Beryllium, dissolved (µg/L as Be)	Boron, dissolved (µg/L as B)	Bromide, dissolved (mg/L as Br)	Cadmium, dissolved (µg/L as Cd)	Chromium, dissolved (µg/L as Cr)	Cobalt, dissolved (µg/L as Co)	Copper, dissolved (µg/L as Cu)
DTX5	01/06/00	1310	2	<2	<2	16	<2	400	0.20	<2.0	0.9	3	6
DTX5	04/11/00	1345	<2	<2	<2	13	<2	330	.19	<2.0	<1.0	3	6
DTX5	07/11/00	1150	<1	<2	<2	13	<2	390	2.1	<2.0	<.8	2	7
DTX5	10/16/00	1015	<2	1	<2	14	<1	470	.25	<.1	<.8	2	7
DTX6	01/06/00	1530	3	<2	E1	9	<2	360	.15	<2.0	<.8	<2	7
DTX6	04/14/00	1000	<2	<1	2	9	<1	340	.17	<1.0	.8	<1	16
DTX6	07/13/00	1300	3	<1	<2	9	<1	340	.23	<1.0	<3.2	<1	9
DTX6	10/11/00	1400	3	<1	<2	8	<1	350	.17	E.1	3.0	1	9
DTX8A	01/13/00	1315	2	<1	<2	22	<1	260	.26	<1.0	E.8	<1	2
DTX8A	04/13/00	1415	<2	<1	<2	21	<1	240	.25	<1.0	<.8	<1	4
DTX8A	07/10/00	1505	<4	<1	<2	19	<1	260	.26	<1.0	<.8	<1	3
DTX8A	10/16/00	1435	3	E.03	<2	17	<1	260	.27	<.1	<.8	1	2
DTX10A	01/11/00	1130	2	<1	<2	16	<1	230	0.25	<1.0	1.5	<1	3
DTX10A	04/17/00	1415	<4	<2	<2	13	<2	240	.26	<2.0	<1.0	<2	6
DTX10A	07/13/00	1020	3	<1	<2	12	<1	250	.25	<1.0	<3.2	1	6
DTX10A	10/18/00	1040	<2	<1	<2	12	<1	250	.28	<.1	<.8	1	5

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[uS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Aluminum, dissolved (µg/L as Al)	Antimony, dissolved (µg/L as Sb)	Arsenic, dissolved (µg/L as As)	Barium, dissolved (µg/L as Ba)	Beryllium, dissolved (µg/L as Be)	Boron, dissolved (µg/L as B)	Bromide, dissolved (mg/L as Br)	Cadmium, dissolved (µg/L as Cd)	Chromium, dissolved (µg/L as Cr)	Cobalt, dissolved (µg/L as Co)	Copper, dissolved (µg/L as Cu)
D6	01/11/00	1355	121	<1	2	<1	<1	890	4.32	<1.0	<4.0	1	15
D6	04/13/00	0945	1	<1	1E9	6	<1	E750	4.32	<1.0	<.8	6	61
D6	07/11/00	1345	<1	<7	E1	<7	<7	820	3.95	<7.0	<.8	<7	35
D6	10/12/00	1300	<1	<1	3	5	<1	970	4.15	.1	<.8	6	33
D13	01/12/00	1005	<1	<1	<2	20	<1	80	.18	<1.0	.9	<1	3
D13	04/17/00	1115	<7	<1	<2	17	<1	80	.17	<1.0	<.8	<1	2
D13	07/07/00	1230	<1	<1	<2	19	<1	100	.15	<1.0	E.5	<1	2
D13	10/16/00	1215	<1	<1	<2	24	<1	110	.19	E.02	<.8	1	3
D17	01/07/00	1500	3	<1	E1	61	<1	60	.08	<1.0	<.8	<1	<1
D17	04/10/00	1415	<1	<1	E1	58	<1	60	.09	<1.0	<.8	<1	<1
D17	07/06/00	1320	<1	<1	E1	56	<1	90	.09	<1.0	<.8	<1	1
D17	10/10/00	1245	<1	<1	E1	59	<1	60	.09	<1	<.8	1	1
D25	01/12/00	1245	4	<1	3	20	<1	300	1.57	<1.0	2.6	4	9
D25	04/10/00	1615	1	<1	3	19	<1	250	1.26	<1.0	<1.0	3	11
D25	07/06/00	1500	<1	<1	3	17	<1	440	10.05	<1.0	<.8	3	10
D25	10/10/00	1455	<2	<2	3	23	<2	400	1.38	<2.0	4.4	5	9

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[uS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; -, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Aluminum, dissolved (µg/L as Al)	Antimony, dissolved (µg/L as Sb)	Arsenic, dissolved (µg/L as As)	Barium, dissolved (µg/L as Ba)	Beryllium, dissolved (µg/L as Be)	Boron, dissolved (µg/L as B)	Bromide, dissolved (mg/L as Br)	Cadmium, dissolved (µg/L as Cd)	Chromium, dissolved (µg/L as Cr)	Cobalt, dissolved (µg/L as Co)	Copper, dissolved (µg/L as Cu)
D29	01/07/00	1130	6	<2	<2	9	<2	180	0.18	<2.0	<1.0	<2	6
D29	04/10/00	1215	3	<1	E1	9	<1	190	.18	<1.0	<1.0	1	10
D29	07/06/00	1105	<1	<1	<2	9	<1	190	.20	<1.0	.8	2	8
D29	10/12/00	1110	<1	<1	<2	8	<1	200	.22	<1.0	<.8	1	7
D30	01/11/00	1600	2	<1	E1	5	<1	490	.72	<1.0	<1	1	5
D30	04/13/00	1200	<2	<1	3	10	<1	440	.70	<1.0	.9	2	18
D30	07/10/00	1245	<13	<1	<2	10	<1	450	.66	<1.0	<.8	2	12
D30	10/12/00	1450	1	<1	E2	10	<1	460	.64	<1.0	<.8	2	9

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[pS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; -, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Iron, dissolved (µg/L as Fe)	Lead, dissolved (µg/L as Pb)	Manganese, dissolved (µg/L as Mn)	Mercury, dissolved (µg/L as Hg)	Molybdenum, dissolved (µg/L as Mo)	Nickel, dissolved (µg/L as Ni)	Selenium, dissolved (µg/L as Se)	Silver, dissolved (µg/L as Ag)	Strontium, dissolved (µg/L as Sr)	Zinc, dissolved (µg/L as Zn)	Uranium, natural (µg/L as U)
DTX1	01/10/00	1345	<10	<2	90	<0.2	6	22	<2	<2	5,960	7	54
DTX1	04/14/00	1220	<30	<1	77	<2	6	4	6	<1	5,340	<14	54
DTX1	07/07/00	0940	<30	<1	96	<2	6	16	E2	<1	5,640	4	48
DTX1	10/17/00	1200	<30	<2	130	<2	7	31	3	<2	5,460	8	50
DTX2	01/10/00	1520	370	<2	3,890	<2	<2	15	<2	<2	4,840	7	36
DTX2	04/14/00	1345	400	<1	3,700	<2	2	2	4	<1	5,000	<10	36
DTX2	07/07/00	1050	460	<1	4,130	<2	2	15	E2	<1	5,450	4	36
DTX2	10/17/00	1005	510	<2	3,640	<2	<2	26	3	<2	5,200	9	34
DTX3	01/10/00	1035	<10	<1	<1	<2	<1	4	8	<1	2,070	2	20
DTX3	04/14/00	1015	<10	<1	<1	<2	<1	6	14	<1	2,770	<3	28
DTX3	07/10/00	1030	<30	<1	<1	<2	<1	9	14	<1	3,070	<5	28
DTX3	10/17/00	1030	<30	<1	E.1	<2	1	2	17	<1	2,910	2	31
DTX4	01/06/00	0955	<10	<2	37	<2	<2	<1	E2	<2	4,180	3	32
DTX4	04/14/00	1145	<30	<2	33	<2	<2	20	3	<2	4,470	<11	35
DTX4	07/14/00	1010	<30	<1	36	<2	<2	10	E2	<2	4,200	3	27
DTX4	10/17/00	1215	<30	<1	69	<2	1	8	12	<2	4,300	5	32

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[pS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day.]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Iron, dissolved (µg/L as Fe)	Lead, dissolved (µg/L as Pb)	Manganese, dissolved (µg/L as Mn)	Mercury, dissolved (µg/L as Hg)	Molybdenum, dissolved (µg/L as Mo)	Nickel, dissolved (µg/L as Ni)	Selenium, dissolved (µg/L as Se)	Silver, dissolved (µg/L as Ag)	Strontium, dissolved (µg/L as Sr)	Zinc, dissolved (µg/L as Zn)	Uranium, natural (µg/L as U)
DTX5	01/06/00	1310	100	<2	157	<0.2	<2	146	2	<2	5,270	3	39
DTX5	04/11/00	1345	120	<2	140	<2	<2	25	<2	<2	5,080	<14	38
DTX5	07/11/00	1150	90	<2	120	<2	<2	10	<2	<2	5,110	3	34
DTX5	10/16/00	1015	160	<1	126	<2	1	9	3	<2	4,540	5	35
DTX6	01/06/00	1530	<10	<2	<2	<2	<2	140	4	<2	5,240	5	37
DTX6	04/11/00	1000	<30	<1	<1	<2	<1	<1	6	<1	5,360	<11	38
DTX6	07/13/00	1300	<30	<1	<1	<2	<1	<1	E2	<1	5,720	5	41
DTX6	10/11/00	1400	<30	<1	1	<2	1	3	4	<2	5,370	7	38
DTX8A	01/13/00	1315	80	<1	180	<2	<1	6	<2	<1	2,460	2	<1
DTX8A	04/13/00	1415	120	<1	181	<2	<1	<1	<2	<1	2,320	<3	<1
DTX8A	07/10/00	1505	170	<1	181	<2	<1	3	<2	<1	2,520	<4	<1
DTX8A	10/16/00	1435	220	<1	172	<2	1	2	<2	<1	2,420	2	<1
DTX10A	01/11/00	1130	3,850	<1	158	<0.2	<1	9	<2	<1	5,450	2	<1
DTX10A	04/17/00	1415	3,970	<2	327	<2	<2	21	<2	<2	5,430	<12	<2
DTX10A	07/13/00	1020	4,130	<1	290	<2	1	3	<2	<1	5,490	4	<1
DTX10A	10/18/00	1040	4,110	<1	294	<2	1	1	<2	<2	5,510	4	M

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[pS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day.]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Iron, dissolved (µg/L as Fe)	Lead, dissolved (µg/L as Pb)	Manganese, dissolved (µg/L as Mn)	Mercury, dissolved (µg/L as Hg)	Molybdenum, dissolved (µg/L as Mo)	Nickel, dissolved (µg/L as Ni)	Selenium, dissolved (µg/L as Se)	Silver, dissolved (µg/L as Ag)	Strontium, dissolved (µg/L as Sr)	Zinc, dissolved (µg/L as Zn)	Uranium, natural (µg/L as U)
D6	07/11/00	1345	<200	<7	3,970	E2	<7	20	9	<7	16,500	42	151
D6	01/11/00	1355	<10	<1	1,571	<2	<1	6	11	<1	15,600	15	121
D6	04/13/00	0945	<500	<1	3,680	<2	3	9	E17	<1	17,500	63	164
D6	10/12/00	1300	<150	<1	3,760	<2	4	8	15	<1	16,500	22	169
D13	01/12/00	1005	20	<1	84	<2	1	10	<2	<1	1,010	2	6
D13	04/17/00	1115	10	<1	82	<2	1	4	<2	<1	968	<3	8
D13	07/07/00	1230	10	<1	76	<2	1	4	<2	<1	997	<1	7
D13	10/16/00	1215	<10	<1	48	<2	1	2	E2	<1	1,160	2	7
D17	01/07/00	1500	<10	<1	249	<2	6	2	8	<1	288	<1	5
D17	04/10/00	1415	<10	<1	213	<2	6	<1	9	<1	305	<1	6
D17	07/06/00	1320	<10	<1	298	<2	6	1	9	<1	340	<1	7
D17	10/10/00	1245	<10	<1	347	<2	6	1	8	<1	295	<1	4
D25	01/12/00	1245	<10	<1	3,100	<0.2	10	38	3	<1	3,230	8	35
D25	04/10/00	1615	<30	<1	3,120	<2	9	14	2	<2	3,200	10	37
D25	07/06/00	1500	<30	<1	3,290	<2	9	33	2	<1	3,370	5	41
D25	10/10/00	1455	<30	<2	3,790	<2	14	14	6	<2	3,550	6	60

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[pS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day.]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Iron, dissolved (µg/L as Fe)	Lead, dissolved (µg/L as Pb)	Manganese, dissolved (µg/L as Mn)	Mercury, dissolved (µg/L as Hg)	Molybdenum, dissolved (µg/L as Mo)	Nickel, dissolved (µg/L as Ni)	Selenium, dissolved (µg/L as Se)	Silver, dissolved (µg/L as Ag)	Strontium, dissolved (µg/L as Sr)	Zinc, dissolved (µg/L as Zn)	Uranium, natural (µg/L as U)
D29	01/07/00	1130	7,210	<2	817	<2	<2	<1	<2	<2	5,830	8	<2
D29	04/10/00	1215	5,970	<1	813	<2	1	7	E2	<2	5,840	19	<1
D29	07/06/00	1105	7,080	<1	788	<2	<1	24	E1	<1	5,810	9	<1
D29	10/12/00	1110	6,640	<1	868	<2	1	7	3	<1	5,740	13	<1
D30	01/11/00	1600	50	<1	149	<2	2	15	<2	<1	5,880	4	17
D30	04/13/00	1200	E40	<1	293	<2	3	2	5	<1	6,010	<16	40
D30	07/10/00	1245	100	<1	253	<2	3	14	<2	<1	5,740	<11	35
D30	10/12/00	1450	1560	<1	260	<2	2	4	3	<1	5,780	6	35

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[aSi/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Plutonium-238, whole water (pCi/L)	Analytical uncertainty, Plutonium-238 (pCi/L)	MDC, Plutonium-238 (pCi/L)	Plutonium-239+240, whole water (pCi/L)	Analytical uncertainty, Plutonium-239+240, (pCi/L)	MDC, Plutonium-239+240 (pCi/L)
DTX1	01/10/00	1345	0	0.012	0.013	-0.002	0.004	0.027
DTX1	04/14/00	1220	--	--	--	--	--	--
DTX1	07/07/00	0940	--	--	--	--	--	--
DTX1	10/17/00	1200	0	0.012	0.014	0.005	0.010	0.014
DTX2	01/10/00	1520	0.008	0.016	0.022	0	0.020	0.022
DTX2	04/14/00	1345	--	--	--	--	--	--
DTX2	07/07/00	1050	--	--	--	--	--	--
DTX2	10/17/00	1005	0	0.007	0.008	0	0.007	0.008
DTX3	01/10/00	1035	0	0.004	0.004	0.001	0.004	0.009
DTX3	04/14/00	1015	--	--	--	--	--	--
DTX3	07/10/00	1030	--	--	--	--	--	--
DTX3	10/17/00	1030	0	0.008	0.009	0.002	0.007	0.018
DTX4	01/06/00	0955	-0.003	0.008	0.027	0.006	0.008	0.008
DTX4	04/14/00	1145	--	--	--	--	--	--
DTX4	07/14/00	1010	--	--	--	--	--	--
DTX4	10/17/00	1215	-0.002	0.003	0.024	0.009	0.012	0.012

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[pCi/cm, picocuries per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Plutonium-238, whole water (pCi/L)	Analytical uncertainty, Plutonium-238 (pCi/L)	MDC, Plutonium-238 (pCi/L)	Plutonium-239+240, whole water (pCi/L)	Analytical uncertainty, Plutonium-239+240, (pCi/L)	MDC, Plutonium-239+240 (pCi/L)
DTX5	01/06/00	1310	0	0.013	0.014	0.003	0.012	0.030
DTX5	04/11/00	1345	--	--	--	--	--	--
DTX5	07/11/00	1150	--	--	--	--	--	--
DTX5	10/16/00	1015	-0.001	0.002	0.016	0.003	0.006	0.008
DTX6	01/06/00	1530	-0.003	0.006	0.042	0	0.018	0.020
DTX6	04/14/00	1000	--	--	--	--	--	--
DTX6	07/13/00	1300	--	--	--	--	--	--
DTX6	10/11/00	1400	0	0.004	0.004	0.002	0.003	0.004
DTX8A	01/13/00	1315	0	0.006	0.006	-0.002	0.003	0.017
DTX8A	04/13/00	1415	--	--	--	--	--	--
DTX8A	07/10/00	1505	--	--	--	--	--	--
DTX8A	10/16/00	1435	0	0.004	0.005	-0.001	0.001	0.010
DTX10A	01/11/00	1130	0.003	0.012	0.030	0.003	0.012	0.030
DTX10A	04/17/00	1415	--	--	--	--	--	--
DTX10A	07/13/00	1020	--	--	--	--	--	--
DTX10A	10/18/00	1040	0.006	0.012	0.022	-0.001	0.010	0.030

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[I_u/cm, micrometers per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Plutonium-238, whole water (pCi/L)	Analytical uncertainty, Plutonium-238 (pCi/L)	MDC, Plutonium-238 (pCi/L)	Plutonium-238+240, whole water (pCi/L)	Analytical uncertainty, Plutonium-238+240 (pCi/L)	MDC, Plutonium-238+240 (pCi/L)
D6	01/11/00	1355	0.005	0.020	0.051	-0.004	0.007	0.051
D6	04/13/00	0945	--	--	--	--	--	--
D6	07/11/00	1345	--	--	--	--	--	--
D6	10/12/00	1300	0	0.007	0.008	0.001	0.007	0.020
D13	01/12/00	1005	-0.001	0.002	0.013	0.000	0.004	0.013
D13	04/17/00	1115	--	--	--	--	--	--
D13	07/07/00	1230	--	--	--	--	--	--
D13	10/16/00	1215	0.001	0.009	0.026	0.002	0.008	0.022
D17	01/07/00	1500	-0.001	0.001	0.008	0	0.004	0.004
D17	04/10/00	1415	--	--	--	--	--	--
D17	07/06/00	1320	--	--	--	--	--	--
D17	10/10/00	1245	-0.001	0.002	0.011	0.002	0.003	0.004
D25	01/12/00	1245	-0.004	0.006	0.034	0.001	0.011	0.034
D25	04/10/00	1615	--	--	--	--	--	--
D25	07/06/00	1500	--	--	--	--	--	--
D25	10/10/00	1455	0	0.008	0.009	0.003	0.006	0.009

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[pS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Plutonium-238, whole water (pCi/L)	Analytical uncertainty, Plutonium-238 (pCi/L)	MDC, Plutonium-238 (pCi/L)	Plutonium-239+240, whole water (pCi/L)	Analytical uncertainty, Plutonium-239+240, (pCi/L)	MDC, Plutonium-239+240 (pCi/L)
D29	01/07/00	1130	-0.002	0.005	0.034	0.006	0.012	0.016
D29	04/10/00	1215	--	--	--	--	--	--
D29	07/06/00	1105	--	--	--	--	--	--
D29	10/12/00	1110	-0.010	0.010	0.051	-0.002	0.005	0.034
D30	01/11/00	1600	0	0.019	0.021	0.008	0.016	0.021
D30	04/13/00	1200	--	--	--	--	--	--
D30	07/10/00	1245	--	--	--	--	--	--
D30	10/12/00	1450	-0.002	0.004	0.031	-0.002	0.004	0.031

¹Value is significantly different from historical or subsequent data at the same site, and analytical bias is suspected. However, insufficient evidence from laboratory to reject or change value.

Table 14. Quality-control data for blank samples associated with ground-water samples collected near Deer Trail, Colorado, 2000

[$\mu\text{S}/\text{cm}$, microsiemens per centimeter at 25 degrees Celsius; mg/L , milligrams per liter; $\mu\text{g}/\text{L}$, micrograms per liter; pCi/L , picocuries per liter; Q , equipment blank; F , field blank; S , source-water blank; $<$, less than; E , value estimated by laboratory; $--$, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC , minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Blank type	Date (mm/dd/yy)	Time (hhmm)	Well number	Specific conductance, lab ($\mu\text{S}/\text{cm}$)	pH, laboratory (standard units)	Calcium, dissolved (mg/L as Ca)	Magnesium, dissolved (mg/L as Mg)	Potassium, dissolved (mg/L as K)	Sodium, dissolved (mg/L as Na)	Acid-neutralizing capacity, titration to pH 4.5 lab (mg/L as CaCO_3)	Chloride, dissolved (mg/L as Cl)
F	01/07/00	1430	D17	4	7.8	<0.02	<0.01	<0.2	<0.1	2	<0.3
F	01/11/00	1400	D6	E1	7.3	<0.02	<0.01	<0.2	<0.1	2	<0.3
F ¹	01/11/00	1405	D6	--	--	<0.01	<0.01	--	<0.1	--	--
F	04/11/00	1315	DTX5	4	7.4	<0.02	<0.01	<0.2	<0.1	2	<0.3
F	04/17/00	1045	D13	E2	8.0	<0.02	<0.01	<0.2	<0.1	2	<0.3
Q	04/21/00	1015	DTX10A	E2	7.5	.08	.02	<0.2	<0.1	2	<0.3
Q ¹	04/21/00	1016	DTX10A	--	--	.07	.01	--	<0.1	--	--
F	07/13/00	1240	DTX6	3	8.4	<0.02	<0.01	<0.2	<0.1	2	<0.3
S ¹	07/14/00	1045	DTX6	--	--	<0.01	<0.01	--	<0.1	--	--
S ¹	07/14/00	1050	None	--	--	<0.01	<0.01	--	<0.1	--	--
F	10/01/00	0940	DTX3	4	7.9	<0.01	<0.01	<0.2	<0.1	2	<0.1
F	10/17/00	1140	DTX1	4	8.1	<0.01	<0.01	<0.2	<0.1	1	<0.1

Table 14. Quality-control data for blank samples associated with ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[µS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; Q, equipment blank; F, field blank; S, source-water blank; <, less than; E, value estimated by laboratory; —, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day.]

Blank type	Date (mm/dd/yyyy)	Fluoride, dissolved (mg/L as F)	Silica, dissolved (mg/L as SiO ₂)	Sulfate, dissolved (mg/L as SO ₄)	Nitrogen ammonia plus organic, dissolved (mg/L as N)	Nitrogen ammonia plus organic, total (mg/L as N)	Nitrogen ammonia, dissolved (mg/L as N)	Nitrite plus nitrate (mg/L as N)	Phosphorus, dissolved (mg/L as P)	Phosphorus, total (mg/L as P)
F	01/07/00	<0.1	<0.1	<0.3	E.10	<0.10	E.021	<0.04	<0.05	<0.05
F	01/11/00	<0.1	<0.1	<0.3	<0.10	<0.10	<0.029	E0.02	<0.05	<0.05
F ¹	01/11/00	—	<.1	—	—	—	—	—	—	—
F	04/11/00	<.1	<.1	<.3	<.10	<.10	<.029	.06	<.05	<.05
F	04/17/00	<.1	<.1	<.3	<.10	<.12	<.029	<.04	<.05	<.05
Q	04/21/00	<.1	E.04	<.3	<.10	<.10	<.029	<.04	<.05	<.05
Q ¹	04/21/00	—	<.1	—	—	—	—	—	—	—
F	07/13/00	<.1	<.1	<.3	<.10	<.10	<.029	<.04	<.05	<.05
S ¹	07/14/00	—	<.1	—	—	—	—	—	—	—
S ¹	07/14/00	—	<.1	—	—	—	—	—	—	—
F	10/11/00	<.2	<.1	E.1	<.10	<.10	<.049	<.04	<.05	<.06
F	10/17/00	<.2	<.1	.2	<.10	<.10	<.049	<.04	<.05	<.06

Table 14. Quality-control data for blank samples associated with ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[µS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCVL, picocuries per liter; F, field blank; S, source-water blank; <, less than; E, value estimated by laboratory; -, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Blank type	Date (mm/dd/yy)	Chromium, dissolved (µg/L as Cr)	Cobalt, dissolved (µg/L as Co)	Copper, dissolved (µg/L as Cu)	Iron, dissolved (µg/L as Fe)	Lead, dissolved (µg/L as Pb)	Manganese, dissolved (µg/L as Mn)	Mercury, dissolved (µg/L as Hg)	Molybdenum, dissolved (µg/L as Mo)	Nickel, dissolved (µg/L as Ni)	Selenium, dissolved (µg/L as Se)
F	01/07/00	<0.8	<1	<1	<10	<1	<1	<0.2	<1	<1	<2
F	01/11/00	<8	<1	<1	<10	<1	<1	<2	<1	<1	<2
F ¹	01/11/00	<2	<1	<1	<3	<1	<1	-	<1	<1	-
F	04/11/00	<8	<1	<1	<10	<1	<1	<2	<1	<1	<2
F	04/17/00	<8	<1	<1	<10	<1	<1	<2	<1	<1	<2
Q	04/21/00	<8	<1	16	E10	<1	<1	<2	<1	<1	<2
Q ¹	04/21/00	<2	<1	16	10	<1	<1	-	<1	<1	-
F	07/13/00	E.6	<1	<1	<10	<1	<1	<2	<1	<1	<2
S ¹	07/14/00	<2	<1	<1	<3	<1	<1	-	<1	<1	-
S ¹	07/14/00	<2	<1	<1	<3	<1	<1	-	<1	<1	-
F	10/11/00	<8	<1	<1	<10	<1	<1	<2	<1	<1	<2
F	10/17/00	<8	<1	E.22	<10	<1	<1	<2	<1	<1	<2

Table 14. Quality-control data for blank samples associated with ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[uS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; Q, equipment blank; F, field blank; S, source-water blank; <, less than; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Blank type	Date (mm/dd/yy)	Solids, residue on evaporation at 180°C, dissolved (mg/L)									
		Aluminum, dissolved (µg/L as Al)	Antimony, dissolved (µg/L as Sb)	Arsenic, dissolved (µg/L as As)	Barium, dissolved (µg/L as Ba)	Beryllium, dissolved (µg/L as Be)	Boron, dissolved (µg/L as B)	Bromide, dissolved (mg/L as Br)	Cadmium, dissolved (µg/L as Cd)		
F	01/07/00	<10	<1	<2.0	<1	<1	<20	<0.01	<1.0		
F	01/11/00	<10	<1	<2.0	<1	<1	<20	<0.01	<1.0		
F ¹	01/11/00	--	<1	--	<1	<1	<2	--	<3		
F	04/11/00	<10	<1	<2.0	<1	<1	<20	<0.01	<1.0		
F	04/17/00	<10	<1	<2.0	<1	<1	<20	<0.01	<1.0		
Q	04/21/00	<10	<1	<2.0	<1	<1	<20	<0.01	<1.0		
Q ¹	04/21/00	--	<1	--	<1	<1	<2	--	<3		
F	07/13/00	<10	<1	<2.0	<1	<1	<20	<0.01	<1.0		
S ¹	07/14/00	--	<1	--	<1	<1	<2	--	<3		
S ¹	07/14/00	--	<1	--	<1	<1	<2	--	<3		
F	10/11/00	<10	<1	<2.0	<1	<1	<10	<0.01	<1		
F	10/17/00	<10	<1	<2.0	<1	<1	<10	<0.01	<1		

Table 14. Quality-control data for blank samples associated with ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[µS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; Q, equipment blank; F, field blank; S, source-water blank; <, less than; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Blank type	Date (mm/dd/yy)	Silver, dissolved (µg/L as Ag)	Strontium, dissolved (µg/L as Sr)	Thallium, dissolved (µg/L)	Zinc, dissolved (µg/L as Zn)	Uranium, natural (µg/L as U)	Plutonium-238 (pCi/L)	Analytical uncertainty, Plutonium-238 (pCi/L)	MDC, Plutonium-238 (pCi/L)	Plutonium-239+240 (pCi/L)	Analytical uncertainty, Plutonium-239+240 (pCi/L)	MDC, Plutonium-239+240 (pCi/L)
F	01/07/00	<1	<1.0	--	<1	<1	0	0.004	0.004	0.001	0.003	0.004
F	01/11/00	<1	<1.0	--	<1	<1	-0.001	0.001	0.008	0	0.004	0.004
F ¹	01/11/00	<1	<1	<1	<1	<1	--	--	--	--	--	--
F	04/11/00	<1	<1.0	--	<1	<1	--	--	--	--	--	--
F	04/17/00	<1	<1.0	--	2	<1	--	--	--	--	--	--
Q	04/21/00	<1	E.8	--	<3	<1	20.012	0.008	0.009	0.001	0.002	0.004
Q ¹	04/21/00	<1	8	<1	1	<1	--	--	--	--	--	--
F	07/13/00	<1	<1.0	--	<1	<1	--	--	--	--	--	--
S ¹	07/14/00	<1	<1	<1	<1	<1	--	--	--	--	--	--
S ¹	07/14/00	<1	<1	<1	<1	<1	--	--	--	--	--	--
F	10/11/00	<1	<8	--	<1	<1	0.003	0.004	0.004	0.002	0.005	0.009
F	10/17/00	<1	<8	--	<1	<1	0.001	0.003	0.009	0	0.004	0.004

¹Blank sample for low-level trace-element analysis.

²Analytical bias is suspected. However, insufficient evidence from laboratory to reject or change value.

Table 15. Comparison of water-quality data for replicate and regular ground-water samples collected near Deer Trail, Colorado, 2000

[RPD, relative percent difference, which is defined as $[(\text{sample value} - \text{replicate value})/(\text{sample value} + \text{replicate value})/2] \times 100$; relative percent difference was calculated on unrounded data; --, not analyzed; ND, not determined because data were less than the minimum reporting limit or estimated; <, less than; E, value estimated by laboratory]

Property or constituent	DTX4				D29				DTX3			
	01/06/00				01/07/00				01/10/00			
	Date	Time	Sample	Replicate	RPD	Sample	Replicate	RPD	Sample	Replicate	RPD	RPD
Specific conductance, lab ($\mu\text{S}/\text{cm}$ at 25°C)			2,950	2,960	0	4,020	4,010	0	1,410	1,410	0	0
pH, WHI laboratory (standard units)			7.0	7.0	0	6.9	6.8	1	7.4	7.4	0	0
Hardness total (mg/L as CaCO_3)			1,700	1,700	0	2,800	2,700	4	720	660	9	9
Calcium, dissolved (mg/L as Ca)			531	523	2	535	526	2	170	158	7	7
Magnesium, dissolved (mg/L as Mg)			86.9	84.0	3	346	327	6	69.9	63.6	9	9
Sodium, dissolved (mg/L as Na)			160	158	1	143	140	2	58.2	54.0	8	8
Potassium, dissolved (mg/L as K)			8.8	8.9	-1	11.1	11.6	-4	6.2	6.0	3	3
Acid neutralizing capacity, titration to 4.5, lab (mg/L as CaCO_3)			353	393	-11	293	292	0	274	274	0	0
Sulfate, dissolved (mg/L as SO_4)			1,590	1,620	-2	2,690	2,710	-1	516	516.86	0	0
Chloride, dissolved (mg/L as Cl)			15.2	15.6	-3	11.5	11.6	-1	17.1	17.2	-1	-1
Fluoride, dissolved (mg/L as F)			2	2	6	5	4	13	4	4	0	0
Bromide, dissolved (mg/L as Br)			22	22	0	18	2	-5	22	22	0	0
Silica, dissolved (mg/L as SiO_2)			11.6	11.8	-1	21.1	20.8	1	15.7	15.4	2	2
Solids, residue on evaporation at 180°C , dissolved (mg/L)			2,800	2,850	-2	4,260	4,270	0	1,090	1,090	0	0
Dissolved solids, sum of constituents (mg/L)			2,620	2,670	-2	3,950	3,930	1	1,030	1,010	2	2
Nitric plus nitrate (mg/L as N)			33	32	3	E.02	E.02	0	2.78	2.78	0	0
Nitrogen ammonia, dissolved (mg/L as N)			.063	.09	-31	.446	.446	0	E.014	<.029	ND	ND
Nitrogen ammonia plus organic, total (mg/L as N)			.29	.30	-2	.51	.50	2	.16	.14	10	10
Nitrogen ammonia plus organic, dissolved (mg/L as N)			.26	.25	6	.45	.48	-6	.19	.14	32	32
Phosphorus, total (mg/L as P)			<.05	<.05	0	E.04	<.05	ND	<.05	<.05	0	0
Phosphorus, dissolved (mg/L as P)			<.05	<.05	0	<.05	E3.04	ND	<.05	<.05	0	0

Table 15. Comparison of water-quality data for replicate and regular ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[RPD, relative percent difference, which is defined as $[(\text{sample value} - \text{replicate value})/(\text{sample value} + \text{replicate value})/2] \times 100$; relative percent difference was calculated on unrounded data; --, not analyzed; ND, not determined because data were less than the minimum reporting limit or estimated; <, less than; E, value estimated by laboratory]

Property or constituent	Well number								
	DTX3			D6					
	Date	04/11/00		04/13/00		D25			
Time	1015	1016	0945	0950	1500	1505			
Sample	Replicate	RPD	Sample	Replicate	RPD	Sample			
Replicate	RPD	Sample	Replicate	RPD	Sample	Replicate			
Specific conductance, lab (µS/cm at 25 °C)	1,880	1,880	0	16,300	16,300	0	4,510	4,490	0
pH, WHI, laboratory (standard units)	7.3	7.3	0	7.2	7.2	0	7.3	7.3	0
Hardness total (mg/L as CaCO ₃)	990	960	3	11,000	10,000	10	2,600	2,500	4
Calcium, dissolved (mg/L as Ca)	240	232	3	446	431	3	638	624	2
Magnesium, dissolved (mg/L as Mg)	94.2	91.1	3	2,320	2,230	4	232	223	4
Sodium, dissolved (mg/L as Na)	72.8	72.3	1	2,100	2,000	5	286	271	5
Potassium, dissolved (mg/L as K)	7.3	7.0	4	12.9	12.6	3	7.3	6.0	20
Acid neutralizing capacity, titration to 4.5, lab (mg/L as CaCO ₃)	314	314	0	640	639	0	496	486	2
Sulfate, dissolved (mg/L as SO ₄)	770	762	1	12,900	13,100	-2	2,520	2,530	0
Chloride, dissolved (mg/L as Cl)	26.0	25.1	4	380	384	-1	99.8	100	0
Fluoride, dissolved (mg/L as F)	.4	.4	0	.8	.7	7	1.0	1.0	3
Bromide, dissolved (mg/L as Br)	.28	.28	0	4.32	4.44	-3	10.05	10.06	-18
Silica, dissolved (mg/L as SiO ₂)	16.6	16.0	4	21.3	21.2	0	28.5	27.7	3
Solids, residue on evaporation at 180°C, dissolved (mg/L)	1,550	1,530	1	20,300	20,800	-2	4,640	4,690	-1
Dissolved solids, sum of constituents (mg/L)	1,440	1,410	2	18,600	18,700	-1	4,130	4,090	1
Nitrite plus nitrate (mg/L as N)	3.80	3.80	0	13.1	12.4	5	3.0	3.0	0
Nitrogen ammonia, dissolved (mg/L as N)	<.029	<.029	0	.045	.059	-27	.138	.148	-7
Nitrogen ammonia plus organic, total (mg/L as N)	.17	.16	7	1.4	1.2	15	.77	.79	-3
Nitrogen ammonia plus organic, dissolved (mg/L as N)	<.10	<.10	0	1.5	1.5	0	.74	.70	6
Phosphorus, total (mg/L as P)	<.05	<.05	0	E.04	E.03	ND	.15	.15	0
Phosphorus, dissolved (mg/L as P)	<.05	<.05	0	E.04	<.05	ND	.15	.15	0

Table 16. Statistical comparison of median concentrations for selected chemical constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000, and lowest applicable water-quality standard

[--, not computed or not applicable ; standard is from Colorado Department of Public Health and Environment (1997); H, health-based standard; A, agricultural standard; E, value estimated by laboratory; *, value estimated because variance was zero]

Well	Sample size for statistical comparison	Minimum ground-water value used for statistical comparison ¹	Maximum ground-water value used for statistical comparison ¹	Median ground-water value used for statistical comparison ¹	Colorado standard	Type of regulatory standard	Probability that the median concentration exceeded the regulatory standard ²
Nitrate ³ (mg/L as N)							
D6	8	11	14.6	12	10	H	0.9961
D13	7	.018	.049	.025	10	H	.0078
D17	8	.679	5.68	2.60	10	H	.0039
D25	7	2.2	7.4	3.4	10	H	.0078
D29	8	.018	.03	.023	10	H	.0039
D30	8	.018	.06	.018	10	H	.0039
DTX1	7	1.1	2.8	1.29	10	H	.0078
DTX10A	8	.018	.025	.018	10	H	.0039
DTX2	8	.018	.05	.022	10	H	.0039
DTX3	8	.018	4.3	3.3	10	H	.0039
DTX4	8	.018	6.52	.241	10	H	.0039
DTX5	8	.018	.1	.018	10	H	.0039
DTX6	8	.22	.32	.25	10	H	.0039
DTX8A	8	.018	.06	.018	10	H	.0039
Arsenic ⁴ (µg/L as As)							
D6	8	1	3.2	2.2	5	H	.0039
D13	8	.05	1	1	5	H	.0039
D17	8	1	2	1.35	5	H	.0039
D25	8	2	6	2.7	5	H	.0352
D29	8	.05	2	1	5	H	.0039
D30	8	.05	2.7	1	5	H	.0039
DTX1	8	1	3.5	1.95	5	H	.0039
DTX10A	8	.05	1	1	5	H	.0039
DTX2	8	.05	2	1.15	5	H	.0039
DTX3	8	.05	1	1	5	H	.0039
DTX4	8	.05	2	1	5	H	.0039
DTX5	8	.05	1	1	5	H	.0039
DTX6	8	.05	2.2	1	5	H	.0039
DTX8A	8	.05	1	1	5	H	.0039
Cadmium ⁵ (µg/L as Cd)							
D6	8	.1	3.5	2	5	H	.0039
D13	8	0.022	0.5	0.5	5	H	.0039
D17	8	.05	.5	.5	5	H	.0039

Table 16. Statistical comparison of median concentrations for selected chemical constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000, and lowest applicable water-quality standard—Continued

[--, not computed or not applicable; standard is from Colorado Department of Public Health and Environment (1997); H, health-based standard; A, agricultural standard; E, value estimated by laboratory; *, value estimated because variance was zero]

Well	Sample size for statistical comparison	Minimum ground-water value used for statistical comparison ¹	Maximum ground-water value used for statistical comparison ¹	Median ground-water value used for statistical comparison ¹	Colorado standard	Type of regulatory standard	Probability that the median concentration exceeded the regulatory standard ²
Cadmium ³ (µg/L as Cd)—Continued							
D25	8	.5	1.5	1	5	H	.0039
D29	8	.038	1	1	5	H	.0039
D30	8	.037	1	.5	5	H	.0039
DTX1	8	.5	1	1	5	H	.0039
DTX10A	8	.05	1	.75	5	H	.0039
DTX2	8	.5	1	1	5	H	.0039
DTX3	8	.05	.5	.5	5	H	.0039
DTX4	8	.1	1	1	5	H	.0039
DTX5	8	.05	1	1	5	H	.0039
DTX6	8	.1	1	1	5	H	.0039
DTX8A	8	.05	.5	.5	5	H	.0039
Chromium (µg/L as Cr)							
D6	8	.4	18	.5	100	H, A	.0039
D13	8	.4	11	.45	100	H, A	.0039
D17	8	.4	3	.4	100	H, A	.0039
D25	8	.4	27	1.55	100	H, A	.0039
D29	8	.4	18	.5	100	H, A	.0039
D30	8	.4	9.5	.95	100	H, A	.0039
DTX1	8	.4	14	.5	100	H, A	.0039
DTX10A	8	.4	5.7	.75	100	H, A	.0039
DTX2	8	.4	11	.7	100	H, A	.0039
DTX3	8	.4	13	.5	100	H, A	.0039
DTX4	8	.4	11	.75	100	H, A	.0039
DTX5	8	.4	8.2	.95	100	H, A	.0039
DTX6	8	.4	11	1.1	100	H, A	.0039
DTX8A	8	.4	.8	.45	100	H, A	.0039

Table 16. Statistical comparison of median concentrations for selected chemical constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000, and lowest applicable water-quality standard—Continued

[--, not computed or not applicable ; standard is from Colorado Department of Public Health and Environment (1997); H, health-based standard; A, agricultural standard; E, value estimated by laboratory; *, value estimated because variance was zero]

Well	Sample size for statistical comparison	Minimum ground-water value used for statistical comparison ¹	Maximum ground-water value used for statistical comparison ¹	Median ground-water value used for statistical comparison ¹	Colorado standard	Type of regulatory standard	Probability that the median concentration exceeded the regulatory standard ²
Copper (µg/L as Cu)							
D6	8	5	61	31	200	A	.0039
D13	8	2	4	2.5	200	A	.0039
D17	8	.05	1	.275	200	A	.0039
D25	8	7	11	9	200	A	.0039
D29	8	6	10	7.5	200	A	.0039
D30	8	5	18	8.5	200	A	.0039
DTX1	8	6	14	8.5	200	A	.0039
DTX10A	8	3	6	5.5	200	A	.0039
DTX2	8	5	11	7	200	A	.0039
DTX3	8	2	5	3	200	A	.0039
DTX4	8	4	8	6	200	A	.0039
DTX5	8	4	8	6.5	200	A	.0039
DTX6	8	6	16	8.5	200	A	.0039
DTX8A	8	2	4	2	200	A	.0039
Lead ⁶ (µg/L as Pb)							
D6	8	.5	3.5	2	50	H	.0039
D13	8	.5	.5	.5	50	H	.0039*
D17	8	.5	.5	.5	50	H	.0039*
D25	8	.5	1.5	1	50	H	.0039
D29	8	.5	1	1	50	H	.0039
D30	8	.5	1	.5	50	H	.0039
DTX1	8	.5	1	1	50	H	.0039
DTX10A	8	.5	1	.75	50	H	.0039
DTX2	8	.5	1	1	50	H	.0039
DTX3	8	.5	.5	.5	50	H	.0039*
DTX4	8	.5	1	1	50	H	.0039
DTX5	8	.5	1	1	50	H	.0039
DTX6	8	.5	1	1	50	H	.0039
DTX8A	8	.5	.5	.5	50	H	.0039*

Table 16. Statistical comparison of median concentrations for selected chemical constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000, and lowest applicable water-quality standard—Continued

[--, not computed or not applicable; standard is from Colorado Department of Public Health and Environment (1997); H, health-based standard; A, agricultural standard; E, value estimated by laboratory; *, value estimated because variance was zero]

Well	Sample size for statistical comparison	Minimum ground-water value used for statistical comparison ¹	Maximum ground-water value used for statistical comparison ¹	Median ground-water value used for statistical comparison ¹	Colorado standard	Type of regulatory standard	Probability that the median concentration exceeded the regulatory standard ²
Mercury (µg/L as Hg)							
D6	8	.05	.2	.1	2	H	.0039
D13	8	.05	.1	.1	2	H	.0039
D17	8	.05	.1	.1	2	H	.0039
D25	8	.05	.1	.1	2	H	.0039
D29	8	.05	.1	.1	2	H	.0039
D30	8	.05	.1	.1	2	H	.0039
DTX1	8	.05	.1	.1	2	H	.0039
DTX10A	8	.05	.1	.1	2	H	.0039
DTX2	8	.05	.1	.1	2	H	.0039
DTX3	8	0.05	0.1	0.1	2	H	.0039
DTX4	8	.05	.1	.1	2	H	.0039
DTX5	8	.05	.1	.1	2	H	.0039
DTX6	8	.05	.1	.1	2	H	.0039
DTX8A	8	.05	.1	.1	2	H	.0039
Molybdenum ⁷ (µg/L as Mo)							
D6	8	.5	4	3.25	.. ⁶	--	--
D13	8	.5	1	1	.. ⁶	--	--
D17	8	6	7	6	.. ⁶	--	--
D25	8	9	14	10.5	.. ⁶	--	--
D29	8	.5	1	1	.. ⁶	--	--
D30	8	1	4	2.5	.. ⁶	--	--
DTX1	8	5	7	5.5	.. ⁶	--	--
DTX10A	8	.5	3	1	.. ⁶	--	--
DTX2	8	1	2	1	.. ⁶	--	--
DTX3	8	.5	1	1	.. ⁶	--	--
DTX4	8	1	1	1	.. ⁶	--	--
DTX5	8	1	1	1	.. ⁶	--	--
DTX6	8	.5	1	1	.. ⁶	--	--
DTX8A	8	.5	2	.5	.. ⁶	--	--

Table 16. Statistical comparison of median concentrations for selected chemical constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000, and lowest applicable water-quality standard—Continued

[--, not computed or not applicable ; standard is from Colorado Department of Public Health and Environment (1997); H, health-based standard; A, agricultural standard; E, value estimated by laboratory; *, value estimated because variance was zero]

Well	Sample size for statistical comparison	Minimum ground-water value used for statistical comparison ¹	Maximum ground-water value used for statistical comparison ¹	Median ground-water value used for statistical comparison ¹	Colorado standard	Type of regulatory standard	Probability that the median concentration exceeded the regulatory standard ²
Nickel (µg/L as Ni)							
D6	8	6	23	12	100	H	.0039
D13	8	2	10	4	100	H	.0039
D17	8	.5	2	1	100	H	.0039
D25	8	4	38	16.5	100	H	.0039
D29	8	.5	24	15	100	H	.0039
D30	8	2	15	12	100	H	.0039
DTX1	8	4	31	16	100	H	.0039
DTX10A	8	1	21	6.5	100	H	.0039
DTX2	8	2	26	14	100	H	.0039
DTX3	8	2	9	3.5	100	H	.0039
DTX4	8	.5	20	12	100	H	.0039
DTX5	8	6	46	11.5	100	H	.0039
DTX6	8	.5	40	6	100	H	.0039
DTX8A	8	0.5	6	3	100	H	.0039
Selenium (µg/L as Se)							
D6	8	6	17	8.4	20	A	.0039
D13	8	.5	1.7	1.1	20	A	.0039
D17	8	7.5	9	8.3	20	A	.0039
D25	8	1	6	2.7	20	A	.0039
D29	8	.5	3.3	1.35	20	A	.0039
D30	8	.5	4.9	1.2	20	A	.0039
DTX1	8	1	5.9	2.55	20	A	.0039
DTX10A	8	.5	2	1.2	20	A	.0039
DTX2	8	.5	4.3	1.55	20	A	.0039
DTX3	8	4	16.5	14.05	20	A	.0039
DTX4	8	1	11.6	2	20	A	.0039
DTX5	8	.5	3.3	1.2	20	A	.0039
DTX6	8	.5	6.1	3.35	20	A	.0039
DTX8A	8	.5	1.2	1.1	20	A	.0039

Table 16. Statistical comparison of median concentrations for selected chemical constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000, and lowest applicable water-quality standard—Continued

[--, not computed or not applicable; standard is from Colorado Department of Public Health and Environment (1997); H, health-based standard; A, agricultural standard; E, value estimated by laboratory; *, value estimated because variance was zero]

Well	Sample size for statistical comparison	Minimum ground-water value used for statistical comparison ¹	Maximum ground-water value used for statistical comparison ¹	Median ground-water value used for statistical comparison ¹	Colorado standard	Type of regulatory standard	Probability that the median concentration exceeded the regulatory standard ²
Zinc (µg/L as Zn)							
D6	8	5	63	30	2,000	A	.0039
D13	8	.5	3	1.75	2,000	A	.0039
D17	8	.5	.5	.5	2,000	A	.0039*
D25	8	5	25	7	2,000	A	.0039
D29	8	8	19	11	2,000	A	.0039
D30	8	4	10	6.5	2,000	A	.0039
DTX1	8	4	8	6	2,000	A	.0039
DTX10A	8	2	10	4.5	2,000	A	.0039
DTX2	8	4	9	6	2,000	A	.0039
DTX3	8	1	2.5	2	2,000	A	.0039
DTX4	8	3	6	4	2,000	A	.0039
DTX5	8	3	7	4	2,000	A	.0039
DTX6	8	5	7	6	2,000	A	.0039
DTX8A	8	1.5	2	2	2,000	A	.0039

¹Many of the values used in the statistical comparison were derived from concentrations that are less than the minimum reporting limit and therefore were set equal to one-half the minimum reporting limit.

²Value is 1 minus the p-value resulting from a one-tailed Sign Test (Helsel and Hirsch, 1995), which is used to indicate the level of statistical evidence that selected constituent concentrations are significantly greater than regulatory standards. A value close to 1.0 indicates more evidence that the median concentration exceeded the standard, whereas a value close to 0 indicates little evidence that the median concentration exceeded the standard. The percent confidence of the test can be determined by subtracting the p-value from 1 and multiplying by 100. For example, if the p-value is 0.100, 1-p is 0.900, so the median concentration is greater than the regulatory standard with 90-percent confidence.

³Data compared to standard are for nitrite plus nitrate. Results indicate nitrite is a minor component.

⁴During 2000, standard was a proposed maximum contaminant level.

⁵All data were less than laboratory minimum reporting limit, except for one value (D6). Minimum reporting limits were sometimes greater than the water-quality standard.

⁶All data were less than laboratory minimum reporting limit. Laboratory minimum reporting limit is less than the water-quality standard.

⁷No regulatory standard for this constituent.

Table 17. Statistical evaluation of monotonic time-series trend using the Kendall's tau correlation coefficient for selected constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000

[Only values in bold indicate statistically significant trends. For this report, the Kendall's tau statistic (Helsel and Hirsch, 1995) is used as an indicator of monotonic correlation between concentration and time. By this method, positive values of Kendall's tau indicate upward trends and negative values indicate downward trends. Kendall's tau is a number between -1 and 1 that indicates increasing strength of the correlation. A p-value is listed to indicate the level of significance of the correlation; --, not computed]

Well	Nitrate		Arsenic		Cadmium		Chromium		Copper		Lead	
	Tau	p-value	Tau	p-value	Tau	p-value	Tau	p-value	Tau	p-value	Tau	p-value
D6	.786	.009	.107	.804	-.428	.174	-.643	.035	.214	.536	-- ¹	-- ¹
D13	.238	.548	.428	.174	-- ¹	-- ¹	-.571	.063	-.036	1.000	-- ¹	-- ¹
D17	.214	.536	-.428	.174	-- ¹	-- ¹	-.607	.048	.714	.019	-- ¹	-- ¹
D25	-.428	.230	-.071	.902	-- ¹	-- ¹	-.250	.458	.250	.458	-- ¹	-- ¹
D29	-- ¹	-- ¹	.071	.902	-- ¹	-- ¹	-.286	.386	-.036	1.000	-- ¹	-- ¹
D30	-.607	.048	.500	.108	-- ¹	-- ¹	-.500	.108	.071	.902	-- ¹	-- ¹
DTX1	.048	1.000	.071	.902	-- ¹	-- ¹	-.571	.063	.428	.173	-- ¹	-- ¹
DTX10A	-- ¹	-- ¹	.357	.266	-- ¹	-- ¹	-.036	1.000	-.178	.621	-- ¹	-- ¹
DTX2	-.393	.216	.142	.710	-- ¹	-- ¹	-.607	.048	.393	.216	-- ¹	-- ¹
DTX3	-.214	.536	-- ¹	-- ¹	-- ¹	-- ¹	-.214	.536	.178	.621	-- ¹	-- ¹
DTX4	.500	.108	.036	1.000	-- ¹	-- ¹	-.286	.386	<.001	1.000	-- ¹	-- ¹
DTX5	-.464	.138	.428	.174	-- ¹	-- ¹	-.750	.013	.071	.902	-- ¹	-- ¹
DTX6	-.250	.458	.464	.138	-- ¹	-- ¹	.071	.902	.214	.536	-- ¹	-- ¹
DTX8A	-.607	.048	-- ¹	-- ¹	-- ¹	-- ¹	-- ¹	-- ¹	.107	.804	-- ¹	-- ¹
Well	Mercury		Molybdenum		Nickel		Selenium		Zinc			
	Tau	p-value	Tau	p-value	Tau	p-value	Tau	p-value	Tau	p-value		
D6	-- ¹	-- ¹	.214	.536	-.107	.804	.464	.138	.071	.902		
D13	-- ¹	-- ¹	.428	.174	.071	.902	-- ¹	-- ¹	-.071	.902		
D17	-- ¹	-- ¹	-.107	.804	<.001	1.000	-.286	.386	-- ¹	-- ¹		
D25	-- ¹	-- ¹	-.250	.458	-.178	.621	.036	1.000	-.250	.458		
D29	.357	.266	-.178	.621	-.464	.138	.464	.138	.107	.804		
D30	-- ¹	-- ¹	<.001	1.000	-.250	.458	.428	.174	-.500	.108		
DTX1	-- ¹	-- ¹	.678	.026	.214	.536	.214	.536	.286	.386		
DTX10A	-- ¹	-- ¹	-.428	.174	-.464	.138	.250	.458	-.357	.266		
DTX2	-- ¹	-- ¹	.286	.386	.143	.710	.464	.138	.071	.902		
DTX3	-- ¹	-- ¹	-.321	.322	.214	.536	.250	.458	.178	.621		
DTX4	-- ¹	-- ¹	<.001	1.000	-.393	.216	.643	.035	-.143	.710		
DTX5	-- ¹	-- ¹	<.001	1.000	-.143	.710	.428	.174	-.036	1.000		
DTX6	-- ¹	-- ¹	-.286	.386	-.321	.322	.250	.458	-.25	.458		
DTX8A	-- ¹	-- ¹	-.250	.458	-.178	.621	-- ¹	-- ¹	-.107	.804		

¹ No concentrations were greater than the laboratory minimum reporting limit.

Table 18. Methods used to analyze streambed-sediment samples collected near Deer Trail, Colorado, 2000

[Samples were analyzed at the National Water Quality Laboratory in Denver, except for constituents footnoted; MRL, minimum reporting level; MDC, minimum detectable concentration (radiochemical samples); ICP, inductively coupled plasma; GFAA, graphite furnace atomic absorption; AA, atomic absorption; ASF, automated segmented-flow spectrophotometry; *, not applicable; mg/kg, milligrams per kilogram; µg/g, micrograms per gram; g/kg, grams per kilogram; pCi/g, picocuries per gram]

Property or Constituent	Units	Analytical method	MRL or MDC
Nutrients			
Nitrogen, ammonia plus organic ¹	mg/kg	Colorimetry, ASF, salicylate-hypochlorite	*
Phosphorus, total as P	mg/kg	Colorimetry, ASF, phosphomolybdate	40
Trace elements			
Aluminum	µg/g	DCP	2.8
Arsenic	µg/g	GFAA	1
Cadmium	µg/g	AA	.1
Chromium	µg/g	AA	.4
Copper	µg/g	AA	2
Lead	µg/g	AA	.1
Mercury	µg/g	AA manual cold vapor	.01
Molybdenum	µg/g	AA	.1
Nickel	µg/g	AA	.1
Selenium	µg/g	AA, Hydride generation, ASF	1
Zinc	µg/g	AA	3.1
Carbons			
Inorganic carbon	g/kg	Coulometric	.2
Organic carbon	g/kg	Wet oxidation	.2
Total carbon	g/kg	Induction furnace (Leco)	.1
Radioactivity			
Gross alpha ²	pCi/g	Thorium-230	*
Gross beta ²	pCi/g	Cesium-137	*
Plutonium-238 ²	pCi/g	Alpha spectrometry	*
Plutonium-239+240 ²	pCi/g	Alpha spectrometry	*

¹ Analyses done by Severn Trent Laboratory in Arvada, Colorado. MRL calculated for each sample.

² Analyses done by Severn Trent Laboratory in Richland, Washington. MDC calculated for each sample.

Table 19. Streambed-sediment data collected near Deer Trail, Colorado, July 17, 2000
 [mg/kg, milligrams per kilogram; µg/g, micrograms per gram; g/kg, grams per kilogram; pCi/g, picocuries per gram; <, less than; ±, plus or minus the analytical uncertainty]

Site	Date	Time	Total nitrogen (mg/kg)	Phos- phorus (mg/kg)	Aluminum (µg/g)	Arsenic (µg/g)	Cadmium (µg/g)	Chromium (µg/g)	Copper (µg/g)	Lead (µg/g)	Mercury (µg/g)	Molyb- denum (µg/g)
Biosolids-applied basin	07/17/00	1500	824	580	8,500	<1	0.11	8.0	12.3	13.1	0.02	0.18
Control basin (no biosolids)	07/17/00	1645	836	540	9,240	<1	.12	10.2	10.5	13.8	.02	.20
Control basin, replicate	07/17/00	1705	730	600	10,400	<1	.14	10.8	12.8	15.4	.02	.21

Site	Date	Time	Nickel (µg/g)	Selenium (µg/g)	Zinc (µg/g)	Carbon, inorganic (g/kg)	Carbon, organic (g/kg)	Gross alpha (pCi/g)	Gross beta (pCi/g)	Plutonium-238 (pCi/g)	Plutonium- 239+240 (pCi/g)
Biosolids-applied basin	07/17/00	1500	10.3	<1	40.8	4.7	7.4	9.8 ± 4.4	44.3 ± 7.7	0 ± 0.002	0.002 ± 0.003
Control basin (no biosolids)	07/17/00	1645	12.5	<1	44.4	4.1	7.2	10.4 ± 4.7	44.2 ± 7.8	0 ± .003	.002 ± .004
Control basin, replicate	07/17/00	1705	12.6	<1	48.0	4.5	8.2	16.9 ± 5.9	44.1 ± 7.6	.001 ± .002	.003 ± .003